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The political consequences of energy price shocks: Evidence from Germany*

Théo Konc[†], Jan Christoph Steckel[‡], Jacob Edenhofer[§], Jens Ewald[¶], Thomas Sterner^{||}

July 31, 2026

Abstract

We study the political consequences of Germany’s 2022–2023 electricity price increases using four waves of original panel data. Identification exploits the German billing system, under which suppliers raise monthly instalment payments at dates plausibly exogenous to household characteristics. In a staggered difference-in-differences design with later-treated households as controls, those facing an above-median increase become 7.5 percentage points more likely to support the far-right *Alternative für Deutschland* (AfD). The shift persists, and attitudes, including support for climate policy, adjust only afterwards. This response emerges even though no climate policy caused the shock, households could not anticipate it, parties at both political extremes opposed the underlying sanctions, and compensation was generous. We interpret our reduced-form results in a spatial voting framework in which the loss raises the value of a party’s promise of lower future energy costs. Our results imply that the standard welfare-economic separation of efficiency from distribution can carry a political cost even when compensation is generous and non-distortionary.

Keywords: Political Backlash, Energy price increase, Right wing populism, Panel Data

JEL Codes: C23, D30, D63, D72, H23, H50, Q40

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1 Introduction

Sudden increases in the cost of living are widely believed to have destabilizing political effects (Guriev and Papaioannou, 2022; Algan et al., 2017). Energy prices, e.g. following geopolitical shocks or climate policy, have attracted growing attention as potential drivers of declining trust in democratic institutions and rising support for populist parties (Voeten, 2024). These concerns have also been prominent in the context of the 2022-2023 energy crisis in Germany, where large and sudden increases in household energy costs coincided with heightened political dissatisfaction and electoral gains for populist parties, particularly the far right populist *Alternative für Deutschland* (AfD). Yet, credible causal evidence on how experienced household price shocks affect political attitudes remains limited. In particular, it is not well understood whether temporary and unexpected price increases can generate persistent political consequences.

This paper studies the political consequences of the 2022-2023 energy crisis in Germany, focusing on household electricity price increases. We collect four waves of panel survey data and estimate the causal effects of increases in individual monthly electricity instalment payments on political attitudes. We find that higher electricity payments have increased the support for the AfD, especially among respondents who have received an above median increase in payments.

Our identification strategy exploits institutional and country-specific features of the German electricity billing system that generate plausibly exogenous and staggered variation in households' exposure to higher electricity costs. Most households receive an annual electricity bill and pay monthly advance instalments based on expected annual costs. Electricity providers can adjust these instalments at any point in time, and households with similar observable characteristics – often even in the same building – may experience payment increases at different moments. As a result, the timing at which households experience an electricity price increase is driven by provider-specific billing practices rather than by household behaviour. We leverage this staggered exposure and use later-treated individuals as a control group in a difference-in-difference framework

with individual fixed effects and estimators robust to heterogeneous and dynamic treatment effects.

We deliberately focus on electricity costs. While the energy crisis affected multiple fuels, we argue that the level of increases in electricity payments were unexpected for many households, while compensation was majorly targeted towards increasing heating costs (Hänsel et al., 2025). We argue that an electricity price increase therefore constitutes a salient economic shock at the individual level.¹ This feature is central to our interpretation of the results, as unexpected shocks are more likely to affect political beliefs rather than merely generating short-lived income effects.

Existing evidence indicates that higher energy costs benefit the radical right. Voeten (2024) finds that Dutch renters directly billed for a legislated gas tax increase became more likely to support the radical right than renters whose utilities were included in their rent, an effect concentrated below median income;² Brännlund and Peterson (2024) document a related pattern across Swedish elections from 2002 to 2018. The settings studied in this literature share three characteristics. The cost was created by an identifiable climate or energy policy, or was publicly attributed to one. Voters anticipated it before they were exposed to the cost.³ The radical right was the most consistent – and in some cases the only – opponent of those policies. The German case differs in all these respects, suggesting that the scope conditions of the findings in the extant literature are broader than thus far recognised.

Our paper contributes to the literature on the political effects of economic shocks by providing causal micro-level evidence on how salient and realised – as opposed to announced or anticipated – price shocks affect the political attitudes of respondents over time. Doing so allows for a more demanding test of the ramifications of energy

¹Survey participants in our survey expected an increase in both, the electricity and the heating bill, see Table 4. An open question asked to households in the first wave of the survey indicated that in particular heating costs increases were expected to be significant. While gas price increases were linked systematically to heating costs, the details of merit-order based electricity billing and hence potential consequences for electricity prices were not mentioned at all.

²The tax was agreed in 2013 and phased in between 2016 and 2020, and every mainstream party supported the package, leaving the radical right as its only vocal opponents.

³The same holds of the German heating law of 2023, whose anticipated costs were debated at length before the law was diluted (Kisting et al., 2026; Bechtel et al., 2026; Kramer et al., 2026).

price shocks. It also contributes to the political economy of energy and climate policy by emphasizing the role of experienced retail price increases, instead of aggregate price movements or policy announcements, and by documenting the backlash against climate policy that price increases unrelated to climate policy can induce, either by directly making people less supportive of climate policy or by indirectly raising their support for anti-climate parties. More broadly, the results speak to the mechanisms through which temporary economic shocks can generate durable shifts in political support, even in the presence of generous compensation.

Overall, our study yields three insights. First, an increase in energy costs can damage climate politics even when no climate policy caused it. Support for climate policy fell among those exposed, and the party that gained opposes decarbonisation. Second, a temporary shock can produce durable political effects. Attitudes adjust after the change in party support, and those later changes reduce a voter’s distance from the party they have joined, even with compensation. Third, economic and political efficiency need not coincide. Relief was paid on fixed administrative dates while exposure arrived on dates determined by suppliers, so equal nominal transfers left households with unequal losses at the moment of exposure. Preserving the price signal and compensating by transfer can therefore be efficient in economic terms while remaining costly in political ones.

The remainder of the paper is organized as follows. Section 2 describes the institutional background of the German energy crisis and the electricity billing system. Section 3 presents the data and empirical strategy. Section 4 reports the main results. Section 5 develops the framework and discusses mechanisms. Section 6 concludes.

2 Context: The German Energy Crisis

2.1 The electricity price shock

Due to its relatively strong reliance on Russian gas and oil imports prior to the war, Germany has been heavily affected by the energy price shock resulting from the Russian invasion to Ukraine. Since 2021, electricity and gas prices have increased by approxi-

mately 30% and 100%, respectively. Although Germany, in line with the entire EU, has managed to starkly reduce its dependence on Russian energy imports and done so at relatively low economic cost (Garicano et al., 2022; Bachmann et al., 2024), the political ramifications of the energy price shock remain unclear and poorly understood, despite strong priors among pundits and commentators. Starting mid 2022 – when the first winter after Russia’s invasion of Ukraine was imminent – government advisors, academics, and journalists warned that the energy price shock may boost the popularity of the right-wing populist *Alternative für Deutschland* (AfD) (Baumgärtner et al., 2022).

Figure 1 provides some correlational evidence supporting this hypothesis – as do the 2023/2024 state elections in Thuringia, Brandenburg, and Saxony, where the AfD performed strongly. Indeed, energy prices increased most strongly for East German households and, at least correlationally, it was among these households that the AfD has gained most support recently (Wehrmann, 2024) – which is especially noteworthy given their already high baseline level of support in the former East.

Although this shock is likely to dissipate over time, with greater diversification in energy sources, stringent climate policies will continue to put the energy market under pressure. Various increases in fossil fuel taxes and carbon levies are planned at both the national and European levels. Perhaps, the most notable examples are the increase in Germany’s national carbon price for the transport and building sectors from €30 to €40 (*Brennstoffemissionshandelsgesetz*) as well as the introduction of the European Emissions Trading scheme for the same two sectors (EU ETS 2) in 2027.

2.2 Energy billing system in Germany

Most German households receive one bill for heating and one for electricity once a year, while paying monthly advance payments (*"Abschlagszahlung"*) on the expected energy costs over the year. Importantly, the instalment payments are defined at the start of the billing period based on the consumption of the previous year and the expected price for the upcoming period.

If the annual sum of the instalment payments is less than the actual costs of heating

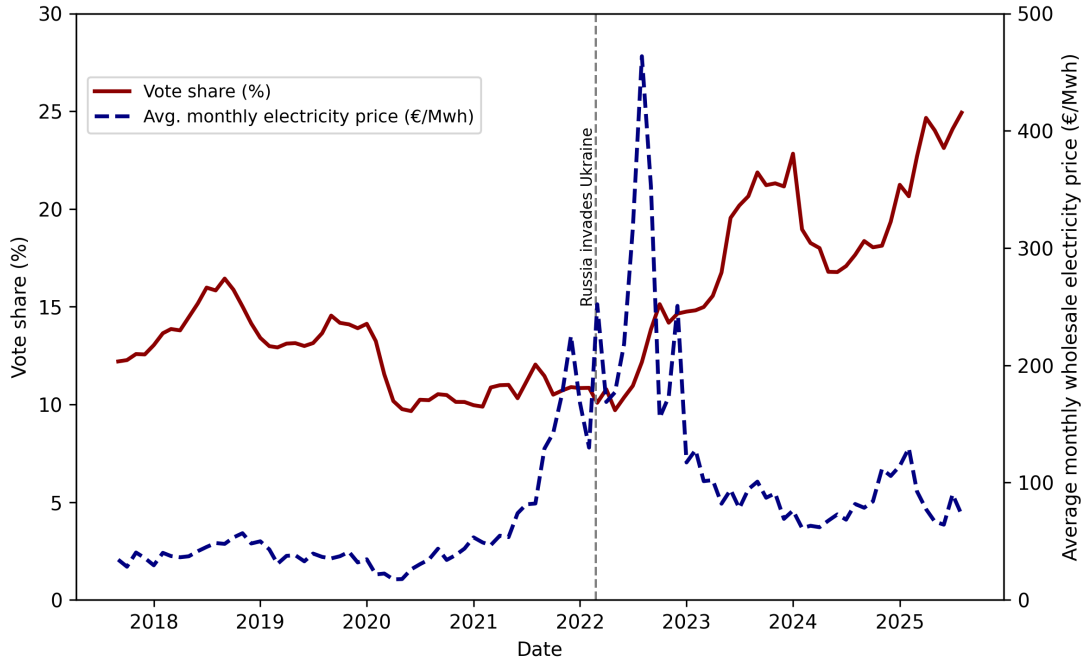


Figure 1: AfD polling and wholesale electricity prices, 2017 to 2025

or electricity (and thus the expected annual bill), households have to pay a corrective bill (*Nachzahlung*). As a consequence, despite increasing electricity spot market prices in mid-2022, households with electricity instalment payments defined earlier during that year kept paying a relatively lower price for their electricity.

These households were exposed to higher retail electricity prices at the start of their next billing period, sometime in 2023. This also means that by December 2023, many households will have received energy corrective bills reflecting the difference between 2022 monthly payments and actual 2022 costs. To reduce the risk of large corrective bills, utilities can also unilaterally adjust (increase or decrease) these advance payments. In the case of an increase, households have the right to quit their contracts ⁴. Such changes can happen at any time, and do not depend on household characteristics.

Another important feature of the German electricity retail market is that households are free to choose utilities. It is common for households in the same street or, even in

⁴In times of high energy price increases, such as the one in 2022 / 2023, it is unlikely that a new contract would leave them better off.

the same building, to be serviced by different utilities, thus creating considerable and quasi-random variation between their monthly bills. The billing system and the absence of live feedback data on consumption given a very low digitization rate in the German energy sector ⁵ makes it difficult to anticipate retail price shocks. In fact, it introduces a disconnect between the current wholesale market and perceived retail prices. This creates a staggered household-level exposure to higher electricity costs, even though the underlying energy shock is common to all households.

2.3 Compensation policies and salience

In response to sharply rising energy costs and inflation, the German government implemented a large set of compensation measures aimed at shielding households from income losses while preserving price incentives. These measures came in three large packages levied at different points in time (*Entlastungspakete*) and included lump-sum transfers, tax reductions and temporary subsidies, with a total fiscal volume of € 95 billion according to the German government. Importantly, most compensation took the form of flat payments of transfers that were not directly linked to households' realized energy consumption, leaving retail prices initially unchanged (Hänsel et al., 2025).

In 2023, the German government intervened in the electricity retail market. Price caps were introduced for electricity (€0.40/kWh), gas (€0.12/kWh), and district heating (€0.095/kWh), applying to 80% of previous year's consumption. Price brakes were also introduced for heating oil and pellets, offering compensation for extreme cost increases. These price brakes applied to the monthly payments starting from March 2023, while the relief for January and February 2023 was paid retroactively in March 2023. The cap on electricity prices curtailed a large part of the increase, albeit not fully. The average retail price peaked in 2023, reaching €0.45/kWh, which represents an increase of 42% after the Russian-Ukraine war (BDEW, 2024). Despite the price cap, electricity retail prices were on average higher in 2023 than in 2022 (DeStatis, 2026). Therefore, the energy shock

⁵Only about 1% of German households are equipped with a smart meter for electricity (Energy transition readiness index, 2023).

had a significant effect on German households budget, despite government compensation packages.

From a standard economic perspective, this policy response closely follows textbook recommendations for addressing price shocks: allow prices to adjust to reflect scarcity while compensating households through non-distortionary transfers. However, the design and compensation packages were complex, involving multiple instruments rolled out over time and administered through different channels. As a result, households’ awareness and understanding of the compensation they received varied substantially (see also Table 4). Survey evidence collected in our data indicates that a large share of respondents did not perceive themselves as having been compensated, despite being eligible for or almost surely receiving transfers.

This imperfect salience of compensation policies is pivotal for our analysis. As transfers were not automatically tied to electricity bills, many households experienced higher electricity payments without immediately associating them with offsetting government support. This disconnect between experienced price increases and perceived compensation contributes to heterogeneity in political responses and play a key role in the belief-updating mechanism discussed in section D.3. To summarise, our objective is to measure the effect of an increase in monthly instalment payments on partisan support and political attitudes, taking advantage of the staggered nature of the treatment.

3 Data and Empirical Strategy

3.1 Data and sample

We collect data from a representative panel of the German population across four survey waves spanning from January 2023 to February 2024. The survey was administered by the professional institute bilendi/respondi. The panel was re-filled to replace respondents who dropped out between waves. Respondents could complete the survey at home or via mobile devices, and they were able to flexibly interrupt and continue the survey.

We conducted the first wave between January 22, 2023 and February 9, 2023, following

a successful pre-test. The survey was sent to 4551 individuals and we retrieved 2004 valid answers. The second wave spanned from 23 May 2023 to 18 July 2023.⁶ It contains 2100 answers, with over 72% of the respondents already present in the first wave. The third wave was conducted between October 10 and November 16. We collected 2096 valid answers, with 61% originating from participants already involved in the initial wave and 77% from those involved in the second wave. We collected the last wave of data between January 26, 2024 and March 11, 2024.⁷ Out of the 2136 complete answers, 50.7% came from respondents involved in all previous waves.

Our main estimation sample consists of all 2276 individuals with two or more observations (N=7173), for which we can in principle measure within-individual variations in electricity payments and political attitudes. As the panel was replenished, we report results for the balanced panel in appendix to account for compositional changes in the unbalanced sample. The balanced panel contains 1083 individuals (N=4332).

The sample was recruited online and is broadly representative of the German population. However, a larger share of middle-sized households, relatively poorer households, and a lower share of individuals over 64 years old indicate a possible selection bias (Table 1). This is a common problem in surveys administered online. Applying weights to the observations is unlikely to solve the issue: individuals over 64 years old who partake in an online survey are unlikely to be representative of all older individuals over 64 years old. Another bias could emerge from a differential attrition rate between the waves. Table 2 shows that the attrition is largely orthogonal to our socio-demographic variables, the energy variables, and the outcome variables which indicates that there is little selection bias into the balanced panel.

The survey is divided into 3 parts (see Appendix A): First, we ask for socio-demographic variable and data on dwelling. Second, we collect data on electricity and heating. Third, we elicit political preferences, including attitudes toward liberal democracy, redistribution, climate policies, xenophobic attitudes and partisan preferences.

⁶75% of the answers were collected before June 15

⁷Over 75% of the data was collected before February 19

Table 1: Representativeness of the sample

	Population	Sample
East Germany	0.149	0.153
Household size=1	0.406	0.305
Household size=2	0.339	0.388
Household size=3	0.121	0.162
Household size $\geq$ 4	0.134	0.144
Under 25 years old	0.045	0.057
Between 25 and 64 years old	0.667	0.798
Older than 64 years	0.289	0.145
Income below 1250 € (Household)	0.133	0.189
Income between 1250 € and 4000 €	0.575	0.561
Income at least 4000 €	0.292	0.250
College degree	0.240	0.297

Note: The data on the population is drawn from the German Federal Statistical Office ([Destatis, 2023](#)). [Destatis \(2023\)](#) addresses the main income earner, which explains some of the differences with our sample.

In the first part, we collect a wide range of socioeconomic and demographic data, including gender, age, education, income, and place of domicile. We also include information on respondents' dwelling, such as surface area, heating source and rental status.

In the second part of the survey, respondents are asked to look at their latest electricity and heating bills, and report their instalment payments, *Abschlagzahlung*, and possible corrective payments, *Nachzahlung*. If they are not able to find it, we ask for estimates. We also include questions on expectations of future energy costs, and on the implementation of energy-saving behaviours by the household (e.g., reducing the temperature in the dwelling, investing in efficient appliances or improved insulation) to approximate efforts to reduce energy consumption.

The data on electricity and heating bills are self-reported, which entails a risk of measurement error. We address this risk in both the survey design and the definition of variables. We ask the respondents to report other data from the bill (date, consumption, etc.), which we use to evaluate the quality of their answer. Respondents reporting a change in instalment payments must indicate the amount they were paying before the change, which we can check using data from previous waves. When constructing the "instalment payment" variables, we exclude all monthly figures greater than €1,000. After

Table 2: Attrition Analysis

	Respondent lost in wave 2, 3 or 4
Socio-demographics	
Female	0.042 (0.047)
Net monthly income of household (€)	-0.021* (0.010)
Unemployed	0.085 (0.080)
University education	-0.048 (0.052)
Electricity and heating	
Live in eastern Germany	-0.019 (0.069)
Gas primary heating source	-0.124 (0.089)
Electricity instalment payments (€)	0.000 (0.000)
Pays extra for renewable electricity	0.056 (0.054)
Expectations	
Expect increase in electricity bill	-0.005 (0.063)
Expect increase in heating bill	0.007 (0.064)
Energy behaviour	
Invest in efficient appliances?	0.043 (0.067)
Invest in efficient heating?	0.063 (0.080)
Reduced car use?	-0.022 (0.055)
Reduce temperature?	-0.001 (0.057)
Invest in improved insulation?	-0.078 (0.086)
Electricity savings?	-0.006 (0.056)
Attitudes	
Trust Government	0.031 (0.054)
Climate change primarily human-induced	0.056 (0.053)
AfD voter	-0.123 (0.081)
Survey is politically neutral	-0.198*** (0.058)
Indices	
Index "Against liberal democracy"	0.006 (0.007)
Index "Pro redistribution"	-0.005 (0.007)
Index "Pro climate policies"	0.001 (0.008)
Index "Xenophobia"	-0.002 (0.005)
R-squared	0.077
Regression coefficient (Robust standard error).	

* : $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$.

A positive coefficient means that an individual with such characteristic is more likely to drop out of the panel, at some point in time.

cleaning the data, we are left with 7,068 observations of electricity instalment-payments and 4,722 observations of heating instalment-payments. To summarise, we observe for each wave whether the respondent’s monthly electricity instalment-payments have increased and the estimated relative change. We describe the treatment measurement in more detail in Section 3.2

Following the energy block, we elicit partisan preferences and four main political attitudes: attitudes toward liberal democracy ⁸, attitudes toward redistribution⁹, attitudes toward climate policies, and xenophobic attitudes¹⁰. We measure the extent to which the respondents agree with specific statements using a 5-point Likert scale. Next, we combine the answers and compute the z-score (see Table 3). The result is an index representing the distance between the attitudes of an individual and the average of the population (Stantcheva, 2023). We use partisan p and the political indices as the outcome variables in our estimation. In particular, we seek to measure whether a positive shock in energy price:

1. Increases support for far-right populists parties (Colantone et al., 2024);
2. Decreases support for price-based climate policies (Drews and van den Bergh, 2016; Douenne and Fabre, 2022);
3. Increases support for redistributive policies, if respondents believe that energy expenditures are very unequally distributed (Martén, 2019; Sommer et al., 2022);
4. Decreases support for liberal democracy (Algan et al., 2017; Guriev and Papaioannou, 2022);
5. Increases agreement with xenophobic statements.

Table 4 provides summary statistics for the balanced panel. We capture a significant variation over time for many variables, for instance when it comes to efforts to reduce

⁸We reproduce questions from the European Value Survey

⁹The questions are adapted from Stantcheva (2021).

¹⁰We combine questions from Mocan and Raschke (2016)’s study on Germany and Hjerm (2005) on Sweden

energy consumption (e.g., by investing in efficient appliances). Here we find a spike in investments in wave 1 when the energy crisis was heavily publicized. In wave 2, we see less such investments which may partly be because they were already done in the first wave and no more additional investments necessary. It may also be a seasonal effect as wave 2 was in the spring rather than winter. We also observe that costs expectations are time-varying. Almost 60% of the sample expected an increase in energy bills in the first wave, while only 36-40% expected an increase in the third wave. This dynamics is likely due to individual receiving their bills and correcting their expectations.

Table 3: Composition of the attitudinal indices

Index		Item type	Content
Against democracy	liberal	Question	There are a variety of opinions about political systems. To what extent do you agree with the following statements?
		Positive contribution	It is best to have strong political leadership without parliamentary constraints; The army should have a greater say in federal decision-making; The federal government should be able to decide without consulting state governments; Political decisions should increasingly be made by expert committees.
		Negative contribution	Democracy has its flaws, but it is better than any other system; The Federal Constitutional Court should continue to review Bundestag legislation.
		Cronbach's α	0.61
Pro redistribution		Question	Would you be for or against an income tax increase for higher earners in the following cases?
		Positive contribution	Revenues used for welfare expansion; industrial investment; military spending; energy subsidies; decarbonisation; reduction of other taxes.
		Cronbach's α	0.67
Pro climate policies		Question	To what extent are you for or against the following climate protection measures?
		Positive contribution	CO ₂ tax across sectors; renewable energy expansion; thermal insulation subsidies; financial support for developing countries.
		Cronbach's α	0.77
Xenophobia		Question	To what extent do you agree with the following statements?
		Positive contribution	Deportation of foreign criminals; preferential treatment for citizens; immigrants should adapt to German traditions; minorities demand special rights.
		Negative contribution	Germany should be multicultural; equal welfare access for immigrants; accept refugees; birthright citizenship; Germany can absorb more immigration.
		Cronbach's α	0.85

Note: Questions translated from German. Original wording provided in the Appendix.

Table 4: Descriptive statistics by waves for the Balanced Panel (1083 households)

Variable	Wave 1	Wave 2	Wave 3	Wave 4
Socio-demographics				
Age	50.575 (14.123)	50.944 (14.138)	51.295 (14.102)	51.541 (14.142)
Female	0.500 (0.500)	0.502 (0.500)	0.502 (0.500)	0.501 (0.500)
Net monthly income	7.448 (2.570)	6.914 (3.498)	7.000 (3.435)	7.098 (3.457)
Dwelling owner	0.448 (0.498)	0.451 (0.498)	0.451 (0.498)	0.452 (0.498)
Live in eastern Germany	0.166 (0.372)	0.166 (0.372)	0.166 (0.372)	0.163 (0.369)
Electricity and heating				
Gas primary heating source	0.571 (0.496)	0.564 (0.496)	0.579 (0.494)	0.589 (0.493)
Electricity instalment payments	107.577 (98.630)	112.438 (106.315)	115.235 (106.933)	118.065 (110.401)
Heating instalment payments	122.544 (129.524)	112.013 (132.747)	109.443 (132.900)	114.935 (126.753)
Electricity from renewables	0.211 (0.408)	0.164 (0.371)	0.159 (0.366)	0.175 (0.380)
Expectations				
Expect increase in electricity bill	0.608 (0.489)	0.392 (0.489)	0.367 (0.482)	0.388 (0.487)
Expect increase in heating bill	0.591 (0.492)	0.416 (0.493)	0.408 (0.492)	0.400 (0.490)
Energy behaviour				
Invest in efficient appliances	0.167 (0.373)	0.127 (0.334)	0.160 (0.367)	0.130 (0.337)
Invest in efficient heating	0.078 (0.269)	0.066 (0.249)	0.078 (0.268)	0.053 (0.223)
Reduced car use	0.266 (0.442)	0.214 (0.410)	0.240 (0.427)	0.239 (0.427)
Reduced temperature	0.535 (0.499)	0.458 (0.498)	0.479 (0.500)	0.446 (0.497)
Improved insulation	0.086 (0.280)	0.059 (0.236)	0.069 (0.254)	0.073 (0.260)
Electricity savings	0.460 (0.499)	0.380 (0.486)	0.429 (0.495)	0.380 (0.486)
Attitudes				
Trust Government	0.307 (0.462)	0.264 (0.441)	0.244 (0.430)	0.215 (0.411)
Climate change primarily human-induced	0.553 (0.497)	0.540 (0.499)	0.531 (0.499)	0.535 (0.499)
AfD voter	0.118 (0.323)	0.137 (0.344)	0.183 (0.387)	0.163 (0.370)
Survey is politically neutral	0.806 (0.396)	0.787 (0.410)	0.801 (0.399)	0.813 (0.390)

¹ Mean (SD)

3.2 Treatment measurement

Our main treatment variable is a binary indicator reflecting whether the respondent has experienced an increase in their electricity instalment payment. The treatment is absorbing and the treatment timing is defined as the first wave in which a respondent reports a higher payment compared to the last observed wave.

Reporting errors might affect the measurement quality of the treatment. As explained in Section 3, we exclude respondents reporting extreme values for their monthly electricity payments ($>€1000$). We also use the answer to a survey question placed right after the energy block and eliciting the perceived quality of answers.¹¹ The main results are estimated on the 'high-quality' sub-sample, consisting of respondents with an above median confidence in their answers (above 83/100 confidence score). We report results using all answers in the appendix.

To account for the possibility that treatment effects are non-linear, we also report results to distinguish the effects of larger and smaller increases. The actual size of the increase is a self-reported measure and might suffer from much larger measurement errors than the binary indicator that the costs have increased. While recent developments in causal inference allow for estimating the effect of continuous treatment effects ([de Chaisemartin and D'Haultfœuille, 2023](#)), our measurement issues would weaken the interpretation of such estimations. Instead, we opt for an exploratory specification, simply distinguishing increases above the sample median and below the sample median. Because of the measurement issues, the results on non-linear effects should be interpreted carefully and constitute descriptive evidence rather than the outcome of a robust causal estimation.

¹¹The english translation of this question is: 'We would like to thank you very much for your time and effort. For our study, it is important to get your assessment of how exactly you were able to answer the questions. Please enter your assessment below on a scale of 1 to 100. "1" means that you are unsure about a lot of answers; "100" means you are very sure of your answers.'

3.3 Identification strategy and assumptions

We use a differences-in-differences design to identify the causal effects of changes in electricity costs on political attitudes. Where we compare outcomes of groups who experienced a change in costs with the groups who did not. Our preferred specification is to use not-yet-treated individuals as control group as it alleviates concerns on self-selection in the treatment, and make full use of the plausibly exogenous staggered arrival of increased in electricity instalment payments.¹²

First let us examine the reasons why staggered exposure to electricity costs increase is plausibly exogenous. While our strategy does not require random timing of the treatment, we need to assume that the timing of the treatment is independent of individual characteristics that could also impact political attitudes. In other words, the key assumption is that the political attitudes of early-treated individuals would have evolved the same way as late-treated individuals, in the absence of a treatment. The key source of variation in our setting is the timing utilities communicated the increase in instalment payments to their customers. We address endogeneity concerns in three ways: first we argue that the features of the German electricity retail market are, by design, consistent with the assumption that treatment timing is exogenous to political trends. Second, we show that untreated, early- and late- treated individuals do not differ systemically in terms of pre-treatment political attitudes. Third, we show that pre-trends are parallel for early- and late-treated individuals, as well as untreated individuals.

3.3.1 Institutional features of German electricity retail market

The unexpected war in Ukraine coupled with the German dependency on Russian gas provides a convincing exogenous shock in energy prices. Gas prices for German households have almost doubled between 2021 and 2023, while electricity prices have increased by 25% over the same period. This shock is global in nature and should in principle affect all households. The lack of natural control group can be mitigated by exploiting the differential timing of the treatment between different groups as in [Li et al. \(2014\)](#).

¹²We report the results using 'Untreated' individuals as a control group in Appendix.

Our identification strategy is to exploit the exogenous variations in energy retail prices across individuals due to the idiosyncrasies of the German energy sector (see Section 2). First, the sector is largely liberalized and diversified. Despite a growing share of renewable power, the energy mix in Germany is still largely dependent on fossil fuels. In 2022, 78% of German primary energy was supplied by natural gas, oil and coal, and only 7% from hydro, wind and solar. The picture is different for electricity generation: 49% came from fossil fuels and 36% from renewable sources [IEA \(2023\)](#). Regions in the North of Germany tend to be less dependent on fossil fuel for electricity generation, while regions in the West are more dependent of gas and coal.¹³. Around 1400 companies operate as electricity suppliers in Germany, and 73% of the local network areas have access to more than 100 suppliers (90% have access to over 50 suppliers). The Herfindahl-Hirschmann Index of the electricity generation market is around 2000, denoting an intermediary level of concentration. The largest company on this market has a 26% market share – compared to 17% in Italy and 79% in France. It should be noted that the largest retail companies, RWE and E.ON, cover all the territory through several regional subsidiaries, but that grid fees are defined at the regional level and have followed heterogeneous trajectories over the last few years. The diversity of electricity generation sources, along with a liberalized market, allows retail suppliers to adopt diverse pricing, investment and long-term contract strategies. An analysis by the "Federal Network Agency" indicates that average price increases between 2021 and 2022 can vary by a factor two depending on the supplier.

Second, German households rarely change suppliers or contracts, even in periods of price fluctuation: 6.4 % have changed contracts and 8.4% have changed suppliers in 2022. These figures include people who have moved home. This reduces the threat that households changed electricity providers strategically in 2022 in order to decrease their exposure to the energy shock.

Third, most German households pay a fixed price over a 12 to 24 months period for

¹³For instance, 90% of the electricity generating capacity of Schleswig-Holstein and Mecklenburg-Schwerin comes from renewable sources, but only 35% in North Rhine-Westphalia. There is however no regional price discrimination in place in Germany

their electricity (*Preisgarantie*), through monthly instalments. Out of over 1400 suppliers, only 52 offered a dynamic pricing contract in 2022, up to 90 in 2023. This means that the spike in wholesale electricity prices in the second semester of 2022 was reflected in higher fixed prices and instalment payments in 2023, when the previous price guaranty expired. Despite a significant decrease in wholesale prices in 2023, the average retail price increased by 25 to 30 %. The timing with which people receive notification that their payments are changing is largely exogenous as it depends on the expiration date of the previous price guaranty, and billing routines of the supplier. In other words, there is no specific period when consumers are supposed to receive a notification that their instalment-payments have increased. Figure 2 shows the share of individuals in each survey wave who have experienced a change in electricity payments.

These characteristics of the German electricity market creates variation in the timing of the changes in electricity retail price and, more importantly, heterogeneous exposure to shocks in the energy market.

Two more threats emerge from the electricity market structure and the timing of the shock: self selection in the treatment group because of the choice of a renewable electricity provider and anticipation of price increases. The choice of renewable electricity providers is a potential source of endogeneity. People selecting these providers could exhibit stronger pro-environmental and left-leaning political attitudes, and received lower price increases, on average. The prices of green electricity providers were about 10% lower than the national average over the 2022-2024 period, largely due to smaller increases in procurement costs. Contrary to our intuition, Table 5 shows no significant link between increases in instalment payments and renewable electricity contracts.¹⁴ We have argued that the timing of *electricity price increases* are exogenous from the household's perspective when controlling for renewable electricity contracts. However, variations in *electricity demand*, and the resulting variation in electricity costs, might be correlated with household characteristics. Even if realised changes in prices are largely random,

¹⁴We still include "renewable electricity provider" as a covariate because of the theoretical source endogeneity it represents.

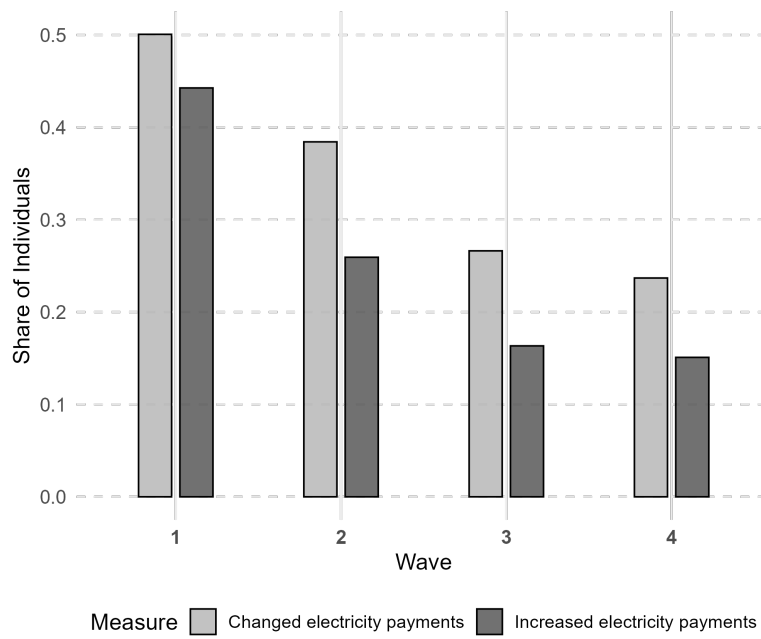


Figure 2: Share of individual per wave with (i) changed electricity payments and (ii) increased electricity payments. Wave 1 covers Jan-Feb 2023, wave 2 May-July 2023, wave 3 October-November 2023 and wave 4 Jan-March 2024.

households expecting a large increase in prices can adapt their behaviours and reduce their energy consumption preemptively. In this case, people informed about the electricity markets who can engage in energy-saving behaviours are likely to experience a lower increase in electricity costs. Indeed, Table 5 shows that variations in electricity costs are negatively correlated with the lag of price expectations. In other words, people who expected a larger price increase in January 2023 experienced a lower increase in electricity bills later that year. To address these two self-selection concerns, we use not-yet-treated individuals as control group in our main specification, and control for "renewable electricity provider" and expectations of price increases measured pre-treatment as a robustness check.

Table 5: Treatment intensity and individual characteristics

	Variation in instalment payments (%)
Expects an increase in electricity bills (lag)	-0.0672*** (0.0175)
Socio-demographics	
Female	0.0075 (0.0176)
East Germany	-0.0091 (0.0251)
Region	0.0028 (0.0021)
Age	0.0010 (0.0006)
Net monthly income of household	0.0006 (0.0028)
Electricity and heating	
Electricity from renewable sources	.00179 (0.0234)
Surface area of the dwelling	-0.0002 (0.0002)
Energy behaviour	
Reported change in energy consumption	-0.0122 (0.0081)
Reduce temperature	-0.0047 (0.0118)
Electricity savings	0.0159 (0.0135)
Invest in efficient appliances	-0.0159 (0.0099)
Invest in improved insulation	-0.0287** (0.0145)
Invest in efficient heating	0.0233* (0.0141)
Reduced car use	-0.0206** (0.0087)
Constant	0.1409** (0.0660)
Observations	2154

Robust standard errors in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

3.3.2 Treatment timing and political attitudes

The institutional features of the German electricity market lead to the plausible assumption that the treatment timing is exogenous. Next, we examine whether the characteristics of early-treated, late-treated, and untreated individual differ systematically.

To do so, we compare the socio-economic and political characteristics measured in the first wave of early-treated individuals (treated in wave 2) and late-treated individuals (treated in waves 3 or 4). For the assumption that treatment timing is exogenous to hold, we should not observe significant pre-treatment differences among individuals treated at different points during the year. Concretely we estimate the following regression:

$$\text{Treated in wave 2}_i = \beta X_{i,1} + \epsilon_i, \quad (1)$$

where $X_{i,1}$ is a vector of individual characteristics measured during the first wave, and "Treated in wave 2" is an indicator valued 1 if the individual is treated between the first and second wave and 0 if treated after the second wave. As our main specification uses not-yet-treated individuals as a control group, we exclude untreated individuals and people treated before the first wave for whom we do not observe a pre-treatment outcome.

Tables 6, 7, and 8 show the outcome of three regressions evaluating the correlation between early treated and individual characteristics. We find that neither socio-economic, political attitudes, nor partisan affiliation can predict the timing of the treatment. The results support our assumption that treatment timing is quasi-random and exogenous to political characteristics.

Table 7 and 8 further support the no anticipation assumption. This assumption is violated if individuals are able to anticipate whether they will be affected by an increase in electricity costs, and adjust their political attitudes accordingly prior to the treatment. For instance, some people might foresee a large increase in costs because they know that their local authorities will not implement relief measures, and reduce their trust in local democracy. In contrast, we find that pre-treatment political attitudes of individuals

treated in wave 2 are similar to those of later-treated individuals.¹⁵

Table 6: Treatment timing and pre-treatment socio-economic characteristics

	Individuals treated in wave 2
Female	-0.126** (0.064)
Age	-0.003 (0.002)
Net monthly income of household (€)	0.005 (0.014)
Unemployed	0.176 (0.110)
University education	0.055 (0.071)
Live in eastern Germany	-0.068 (0.085)
Dwelling surface	0.000 (0.001)
House owner	0.113 (0.076)
Observations	257
Robust standard errors in parentheses.	
* p<0.10, ** p<0.05, *** p<0.01	

Table 7: Treatment timing and pre-treatment political attitudes

	Individuals treated in wave 2
Index trust in political institutions	-0.032 (0.042)
Index Populism	-0.003 (0.036)
Index pro redistribution	0.022 (0.036)
Index pro climate policies	-0.049 (0.041)
Index xenophobia	-0.031 (0.040)
Observations	229
Robust standard errors in parentheses.	
* p<0.10, ** p<0.05, *** p<0.01	

¹⁵This is also the case for individuals who received a higher than median increase in instalment payments, see Tables 16, 17 and 18 in Appendix B. Tables 13, 14 and 15 further show that early-treated and never-treated individuals have similar baseline characteristics.

Table 8: Treatment timing and pre-treatment partisan affiliation

	Individuals treated in wave 2
RWPs support (AfD)	-0.076 (0.146)
Social-democrats support (SPD)	0.020 (0.143)
Conservatives support (CDU)	-0.060 (0.139)
Greens support (Grünen)	-0.072 (0.144)
Liberals support (FDP)	0.146 (0.185)
Radical left support (Die Linke)	-0.312* (0.171)
No party affiliation	-0.217 (0.139)
Observations.	262

Robust standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

3.3.3 Parallel trends and no anticipation

We need two supplementary assumptions: no anticipation and parallel trends in outcomes.

The no anticipation assumption can be written:

$$Y_{i,t,g}(1) - Y_{i,t}(0) = 0 \quad \forall t < g \quad (2)$$

This assumption is fulfilled if the attitudes measured prior to the treatment are not influenced by expected increases in electricity costs. It is violated if individuals are able to anticipate whether they will be affected by an increase in electricity costs, and adjust their political attitudes accordingly. For instance, some people might foresee a large increase in costs because they know that their local authorities will not implement relief measures, and reduce their trust in local democracy. Similarly to 3.2, the "no anticipation" assumption is likely to hold when estimating the treatment effect conditional on price expectations. We test if the outcome at time t is different between the group treated at $t + 1$ and the control group (individual who will be treated at $t + 2$ or $t + 3$). Formally, we estimate the following regression excluding already treated individuals:

$$Y_{i,t} = \beta \text{Treated next period}_{i,t} + \tau_t + \epsilon_{i,t}, \quad (3)$$

where $Y_{i,t}$ is the political attitude or partisan affiliation of individual i at time t and 'Treated next period $_t$ ' is an indicator reflecting whether the individual will be treated at $t + 1$.

Tables 9 and 10 show no effect of the treatment on the prior outcomes, in particular concerning support for populists. Tables 19 , 20, 21 and 21 in Appendix B show that the no anticipation assumption is also likely to hold taking the never-treated individuals as control group, and for individuals experiencing a high increase in electricity instalment.

Table 9: Pre-test of no anticipation on partisan affiliation

Partisan Affiliation	Estimate (S.E)	p value	Observations
RWPs support (AfD)	-0.002 (0.033)	0.962	495
Social-democrats support (SPD)	0.065 (0.037)	0.075	495
Conservatives support (CDU)	0.030 (0.040)	0.454	495
Greens support (Grünen)	0.022 (0.034)	0.516	495
Liberals support (FDP)	0.013 (0.023)	0.576	495
Radical left support (Die Linke)	-0.055 (0.030)	0.066	495
No party affiliation	-0.059 (0.044)	0.177	495

Table 10: Pre-test of no anticipation on political attitudes

Political Attitudes	Estimate (S.E)	p value	Observations
Index trust in political institutions	0.019 (0.107)	0.857	484
Index populism	0.037 (0.097)	0.706	465
Index pro redistribution	0.157 (0.101)	0.120	453
Index pro climate policies	0.008 (0.108)	0.943	468
Index xenophobia	0.000 (0.105)	0.999	475

For staggered treatment, the parallel trend assumption can be written as:

$$\mathbb{E}(Y_{i,g,t}(0) - Y_{i,g,t-1}(0)|G_i = g, B_i = 1) - \mathbb{E}(Y_{i,g,t}(0) - Y_{i,t-1}(0)|G_i = g) = 0, \quad (4)$$

where G_i denotes the first wave in which individual i experienced an increase in prices. With this parallel trends assumption, we impose that the average change in attitude of the untreated individual is the same as the change in attitude of the potential counterfactual individuals in every treatment timing group.

We test our parallel trends assumption using pre-treatment trends and post treatment trends. We lack pre-treatment data for individuals treated before the second wave, hence we compare the pre-treatment trend in outcomes between waves 1 and 2 for individuals treated in wave 3 and wave 4:

$$Y_{i,2} - Y_{i,1} = \beta \text{ Treated in period } 3_i + \epsilon_i, \quad (5)$$

The results are displayed in Tables 11 and 12. They show no significant differences in pre-treatment trends. Tables 23 , 24, 26 and 26 in Appendix B show that the parallel trends assumption is also likely to hold taking the never-treated individuals as control group, and for individuals experiencing a high increase in electricity instalment.

Table 11: Pre-test of parallel trends for individuals treated at $t = 3$ vs. $t = 4$

$\Delta_{2,1}$: Political Attitudes	Estimate (S.E)	p value	Observations
Index trust in political institutions	-0.067 (0.126)	0.592	202
Index Populism	-0.218 (0.142)	0.128	192
Index pro redistribution	0.510 (0.232)	0.031	184
Index pro climate policies	0.108 (0.117)	0.358	194
Index xenophobia	-0.065 (0.136)	0.634	200

Table 12: Pre-test of parallel trends for individuals treated at $t = 3$ vs. $t = 4$

$\Delta_{2,1}$: Partisan Affiliation	Estimate (S.E)	p value	Observations
RWPs support (AfD)	-0.041 (0.030)	0.172	208
Social-democrats support (SPD)	0.041 (0.043)	0.343	208
Conservatives support (CDU)	0.056 (0.033)	0.095	208
Greens support (Grünen)	-0.077 (0.046)	0.095	208
Liberals support (FDP)	0.000 (NA)	NA	208
Radical left support (Die Linke)	-0.015 (0.015)	0.321	208
No party affiliation	0.021 (0.048)	0.670	208

3.4 Estimation strategy

We are interested in identifying the immediate average effect of being exposed to an increase in electricity instalment payments on political attitudes. We measure the effect

of the increase in the first survey wave after that increase is reported (see Figures 14 and 15 in Appendix for estimates of dynamic effects).

Let D_{it} denote the absorbing treatment indicator and let

$$G_i = \min\{t : D_{it} = 1\} \quad (6)$$

denote the first wave in which respondent i reports an increase. All respondents in the estimation sample are untreated in their first observed wave. For respondents first treated in wave g , we define the cohort-specific effect at the time of the treatment as

$$ATT_{g,0} = \mathbb{E}[Y_{i,g}(1) - Y_{i,g}(0) \mid G_i = g], \quad (7)$$

Here, $Y_{i,g}(1)$ is the potential outcome under the observed treatment path in which the respondent reports an increase by wave g , while $Y_{i,g}(0)$ is the counterfactual outcome under a treatment path in which the respondent *has not yet experienced* an increase by that wave.

We use the estimator by [de Chaisemartin et al. \(2022\)](#). For each treatment cohort, it compares the change in outcomes between waves $g - 1$ and g among respondents first treated in wave g with the contemporaneous change among respondents who remain untreated in wave g :

$$DID_{g,0} = \mathbb{E}[Y_{i,g} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{i,g} - Y_{i,g-1} \mid G_i > g], \quad (8)$$

Respondents who are treated later can therefore serve as controls only before their own treatment begins.¹⁶

In our design, the effect is identified from respondents first treated in waves 2 and 3. Respondents first treated in wave 4 can contribute to earlier control groups but do not contribute as treated observations because there are no later-treated individuals in wave

¹⁶The estimand reported in Section 3.2 is the weighted average of these cohort-specific effects, see [de Chaisemartin and D'Haultfoeulle \(2023\)](#) and R package: `did_multiplegt_dyn`.

4.¹⁷ If the assumptions discussed in Section 3.3 hold, $DID_{g,0}$ identifies $ATT_{g,0}$.

To account for the fact that the effect of higher energy bills might depend on the relative magnitude of the increase, we also measure the treatment effect of experiencing an above median increase in costs. In this, treatment begins when a respondent first reports an increase exceeding the sample median of 31%.

4 Results

4.1 Effects of electricity price shock on voting intention

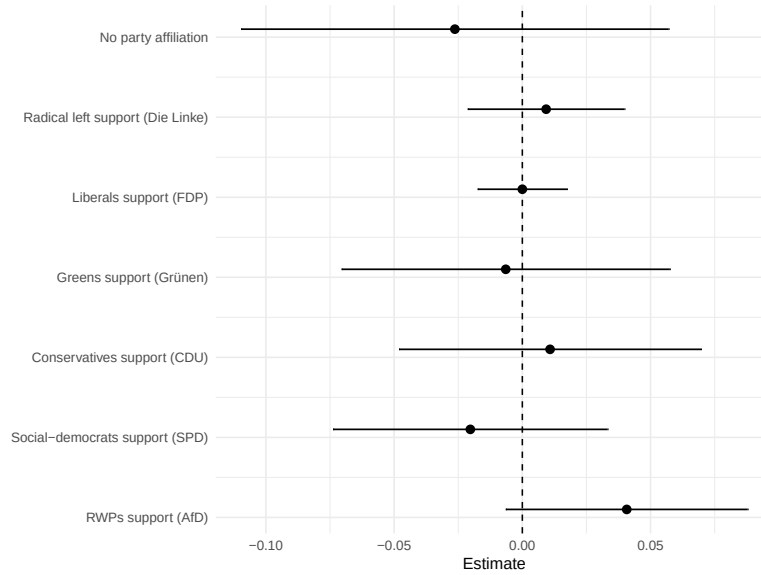
We find that experiencing higher electricity costs has an effect on partisan support of the right wing populist party, AfD, especially among households with an above-median increase. The ATT for any increase in instalment payments is generally not significantly different from 0 (Figure 3, Panel a), while the ATT for an above-median increase is large and significant (Figure 3, Panel b).¹⁸ Households experiencing a large increase are 7.5 percentage points more likely to support the right-wing-populist party (AfD). Support for other parties are not significantly different from zero. Notably, though, support for the Green party is largely reduced, particularly for people that have been affected by higher than median increases.

We perform a series of robustness checks (see Appendix C). Figure 4 summarises the main learnings for the 'above-median increase' treatment. The point estimate remains positive when using never-switchers as controls, adding baseline socio-economic characteristics interacted with survey wave, separating electricity increases with and without a simultaneous heating-payment increase, restricting the analysis to internally consistent bill reports (self-reported values are consistent within a €10 margin across waves), and clustering the standard errors at the federal-state level.

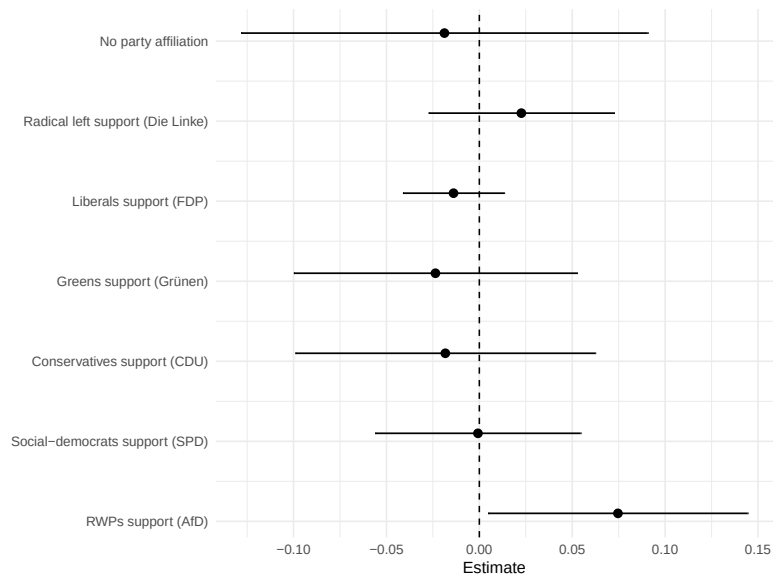
¹⁷Our sample contains 160 eligible switchers, with 72 above-median switchers.

¹⁸Standard-errors are clustered at the individual level

Figure 3: Unconditional D-i-D effects of receiving a higher electricity bill on partisan affiliation.

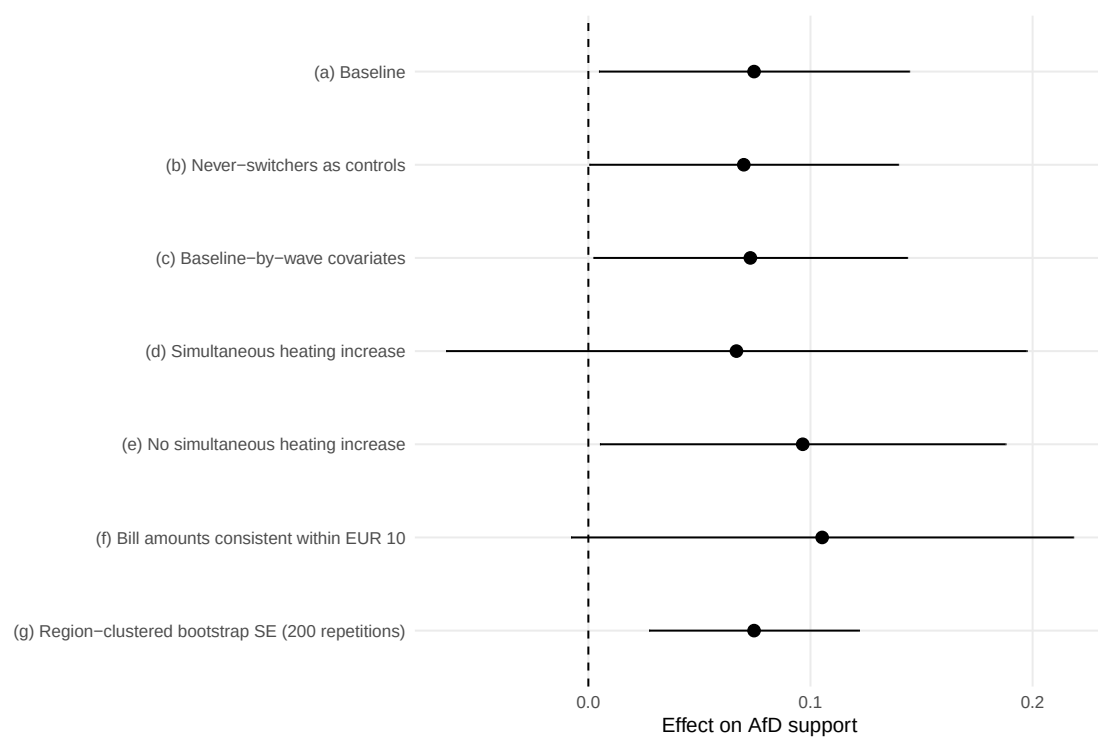


(a) ATT of the binary treatment: 'Increased electricity costs'



(b) ATT of the binary treatment: 'Above median increase in electricity costs'

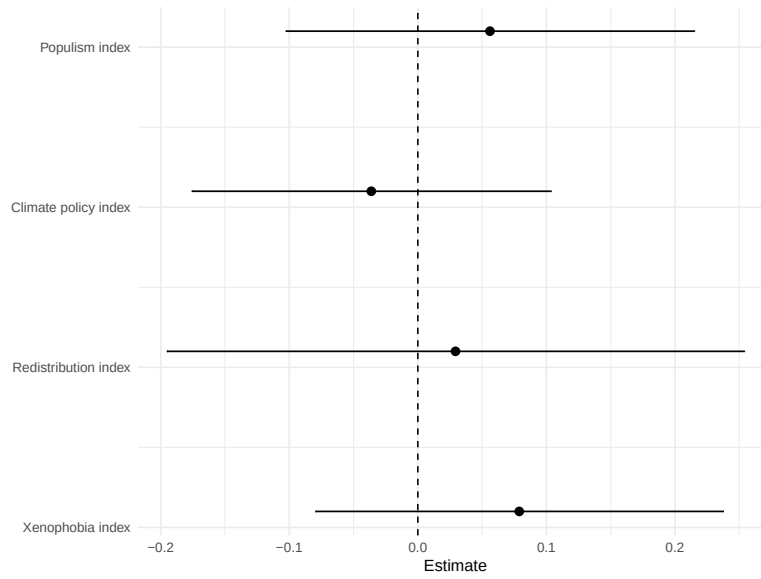
Figure 4: Robustness checks of the D-i-D effect on support for the RWP party (AfD) of the binary treatment: 'Above median increase in electricity costs'



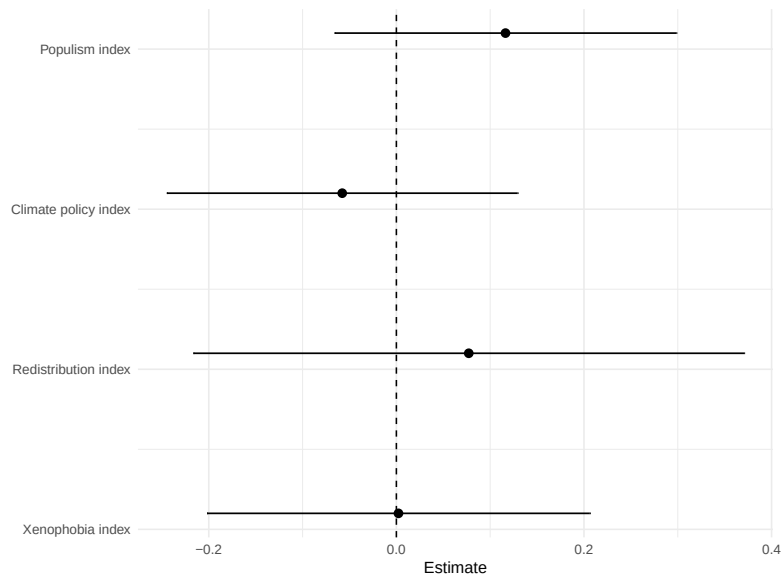
4.2 Effects of electricity price shock on attitudes

To provide further evidence on the political effects of the energy shock, we measure the changes in political attitudes of respondents receiving an increase in electricity costs (Figure 5). In particular, we measure changes in attitudes towards populism, climate policy, redistribution and xenophobia. We find that - on average - effects on political attitudes are small and very imprecisely estimated. In particular, we find no effect on support for redistribution nor on populists attitudes. However, when looking into how effects change over time, that is, whether effects change when comparing different waves, we indeed find indication for a strengthening effect for populism for individuals that have been treated higher than the median (see Figure 15). Those effects are positive and statistically significant from zero.

Figure 5: Unconditional D-i-D effects of receiving a higher electricity bill on political attitudes.



(a) ATT of the binary treatment: 'Increased electricity costs'



(b) ATT of the binary treatment: 'Above median increase in electricity costs'

5 Discussion

We find that experiencing large increases in electricity costs raises support for the far right AfD, and that this shift persists over time. However, our analysis thus far sheds no light on the mechanism underlying these findings – why a material shock of this kind moves voters toward a right-wing populist party specifically. In this section, we develop an interpretation based on a simple framework in which voters weigh ideological distance from a party – how closely its positions across issues like immigration, redistribution, and climate match the voter’s own – against the material costs its policies imply, and in which a shock lowers the threshold at which a materially attractive but more distant party becomes preferable. We then discuss whether the shift in party support is accompanied by lasting changes in political attitudes. Finally, we discuss the role of common explanations for mechanisms, including redistributive preferences moderated by attitudes to Ukraine, energy poverty, welfare chauvinism, and attributional learning about government competence – against pre-treatment moderators and split-sample estimates¹⁹. Finally, we discuss scope conditions and broader implications of our analysis.

5.1 From temporally idiosyncratic shock exposure to party choice

To understand how the electricity price shock affects party choice, it is useful to distinguish two respects in which a voter evaluates the parties in their consideration set. The first is spatial proximity. Each voter holds an ideal point in a multidimensional policy space that spans issues, such as immigration, redistribution, climate policy, foreign policy, and the defence of democratic institutions, and each party occupies a position in that space, following the standard multidimensional spatial voting model ([Downs, 1957](#); [Enelow and Hinich, 1984](#)). A party is spatially close to a voter when it holds the positions the voter prefers on the issues the voter weights most heavily, and distant when it diverges from them on these issues. The second is material. Parties’ programmes imply

¹⁹Our analysis in this section is subject to two limitations: subgroup comparisons and post-treatment attitudes do not by themselves identify causal mechanisms (e.g. [Blackwell et al., 2024](#); [Kwon and Roth, 2026](#)), and what follows is our own rationalisation of the patterns rather than a test of pre-registered relationships, offered in the spirit of narrowing plausible explanations ([Spirling and Stewart, 2025](#)).

different future energy costs for households, and the voter forms an expectation of these costs, net of any compensation, under each party. The material loss from higher electricity payments can therefore change a voter’s party choice through the second channel alone, without any change in underlying ideological positions.

The material effects of the shock on party choice become clear once we combine the diminishing marginal utility of income with the shock’s reduction of the voter’s disposable income. The loss consequently raises the marginal utility of money, thereby making any prospective reduction in future energy costs more valuable. If a party promised lower future energy costs more credibly than the party the voter currently supports, the shock increases that party’s material advantage. The voter switches once this advantage exceeds the additional spatial distance they must accept in supporting the alternative. Formally, a voter switches from her current party j to the AfD when

$$u(R_i - z_i - q_{iA}) - u(R_i - z_i - q_{ij}) \geq D_{iA} - D_{ij}, \quad (9)$$

where z_i is the household’s net loss from the instalment increase, q_{ij} the expected present-value energy cost under party j , and D_{ij} the voter’s spatial distance from party j (full derivation in Appendix F.1). This predicts a threshold rather than a uniform response: a small loss leaves the material advantage below the spatial gap for most voters, while a large loss clears it for those already closer to the margin – consistent with our finding that only above-median increases in instalment payments move support toward the AfD.

The former explains which parties voters become less likely to support. A price shock can reduce support for the parties in government during the shock (Duch and Stevenson, 2008; Hobolt and Høyland, 2011; Margalit, 2019), lowering a governing party’s appeal or removing it from the voter’s consideration set entirely for those who could have imagined voting for a governing party before the shock. However, it cannot account to which party a voter moves to. This, in turn, depends on the spatial-material trade-off set out above, applied to the remaining alternatives in the voter’s consideration set (Enelow and Hinich, 1984). During most of the study period, the AfD most clearly linked lower

energy costs to weaker sanctions on Russia and continued imports of Russian energy.²⁰ Voters who found this link credible, and whose broader positions - most prominently on migration – already placed the AfD within spatial reach had the clearest material reason to switch. Voters far from the AfD on the issues they weight most heavily would not move, independent of the promised relief.

The argument yields two implications for compensation policies. First, its timing relative to exposure matters. A transfer paid on a fixed date can precede exposure for one household and follow it for another, so equal nominal transfers can leave households with unequal losses at the moment of exposure. Compensation arriving close to the first higher instalment should therefore attenuate the political response when households cannot smooth consumption without cost. Figures 10-13 in the Appendix examine this descriptively, though the distance between exposure and compensation is not independently assigned.

Second, the efficacy of compensation in preventing switches to anti-system parties declines in a voter’s spatial distance from the mainstream parties. Compensation shrinks the material advantage of the anti-system alternative, which retains a voter only when that advantage would otherwise exceed the spatial distance they must accept in switching. Voters already close to the anti-system party need only a small material advantage to switch, so compensation does least to hold precisely those voters most at risk of defection. Compensation is therefore most effective when it targets households close to the point of exposure and where voters remain close to the mainstream parties, and least effective where the anti-system party is already within reach.

5.2 From party choice to attitudinal spillovers

The previous subsection explains a change in party choice that need not involve any prior change in political attitudes. Voters switch on the dimensions most salient to them

²⁰Evidence from the special edition of the Chapel Hill Expert Survey presented in the appendix indicates that the AfD was among the German parties most opposed to sanctions and most supportive of continued energy imports (Polk et al., 2024). The BSW later adopted a similar position. Its later entry and lower electoral viability made it a less established alternative during most of the study period.

while tolerating disagreement on less salient ones (Diermeier and Schnabel, 2025; Velez et al., 2026). We use the term attitudinal spillovers for changes in a voter’s positions on these other, less salient dimensions, arising after the switch and as a consequence of it. Such spillovers matter because they determine whether a temporary material shock has lasting political consequences. If a voter who switched on energy costs alone later moves towards the party on immigration, climate, or redistribution, the switch acquires reasons of its own and can outlast the material burden that first produced it. A transient shock would then translate into a lasting realignment instead of a passing protest.

The panel data exhibit a pattern consistent with such spillovers. Appendix D.2 shows that new AfD supporters already resembled established supporters in their xenophobic attitudes before the switch, whereas their support for climate policy declined only afterwards, between waves 3 and 4. The corresponding change in redistribution is sizeable but estimated with less precision, and estimates for the broader attitude indices have wide confidence intervals. These comparisons show that the temporal order that spillovers require is plausible, namely stability on the dimensions that drove the switch and movement on others afterwards. They do not, however, identify a causal effect of AfD support on subsequent attitudes.

Several mechanisms can engender this pattern, which we cannot disentangle with our data. The immediate loss can directly reduce support for climate policies that voters expect to raise household expenditure, with no role for the party. Such voters may continue to value lower emissions while opposing carbon taxes, energy levies, or public expenditure on decarbonisation. Alternatively, the movement may be a genuine spillover from the switch. It could work through closer attention to the party’s elite cues (Slothuus and Bisgaard, 2021; Barber and Pope, 2024), through the propagation of a change across an interconnected web of beliefs (Converse, 1964), or through the adjustment of remaining attitudes to reduce the cognitive dissonance of disagreeing with one’s new party (Callander and Carbajal, 2022). Redistribution is subject to the same ambiguity. In our analysis, the index combines support for taxing high earners with proposed uses of the revenue, including social assistance, industrial investment, military expenditure,

subsidies for energy, decarbonisation, and reductions in other taxes. Movement in it can reflect a reconsideration of material costs, a spillover, or both. Because attitudes move only after the switch and the mechanisms overlap, we cannot establish which of them operates, and we leave this to future research. Whatever their source, these later changes reduce a voter’s spatial distance from the AfD and can therefore sustain the initial electoral response after the material burden declines.

5.3 Other mechanisms and empirical limitations

We report a broader range of analyses in Appendix D.3 concerning awareness of compensation, perceived waste in government, support for Ukraine, welfare chauvinism, income, and control over energy use. These analyses have limited statistical power because dividing the sample produces large standard errors and a low probability of detecting plausible differences in effects. Their confidence intervals remain compatible with effects of meaningful size. We therefore find no strong evidence for these relationships. The corresponding mechanisms may nevertheless be present ([Arel-Bundock et al., 2026](#)).

While we acknowledge these limitations, we emphasise two points. First, some proposed mechanisms can operate in conjunction with the interpretation developed above. Beliefs about waste can reduce the expected value of compensation, beliefs about sanctions can affect expected energy costs, and perceptions of government competence can intensify retrospective sanctioning. Changes in climate policy or redistribution may reflect a direct response to material loss, attention to party cues, or both.

Second, alternative explanations should be evaluated by whether it accounts for the joint distribution of the empirical patterns, not just any given individual pattern. The electoral response is clearest after larger instalment increases and in support for the AfD, although estimates for other parties are imprecise. New AfD supporters resembled established supporters in their xenophobic attitudes before switching, movement after the switch is clearest on climate policy, and the increase in AfD support persists. An account based only on incumbent punishment explains withdrawal from governing parties without determining which opposition party benefits. A general cultural conversion account must

explain why similarity in xenophobic attitudes preceded the switch and why subsequent movement is clearest on climate policy. An account based on redistribution must explain why the clear electoral response appears in support for the AfD. None of these accounts is excluded. The interpretation developed above connects the sequence of outcomes while remaining compatible with complementary processes.

5.4 Scope conditions and broader implications

While the interpretation developed above rationalises our findings, it raises the question of its scope conditions. The first concerns what distinguishes an electricity price shock from shocks to other goods or to a household's labour market position. The second concerns what distinguishes the German case from other settings in which electricity prices rose sharply.

The first scope condition concerns the nature of the good and, concomitantly, the nature of the price shock. Electricity is a necessity. Households cannot store a meaningful quantity, ordinary domestic activity requires a minimum level of consumption, and the capacity to adjust in the short run is limited and uneven.²¹ Investment solar panels or efficient appliances can reduce future use, although these adjustments require substantial capital and take time (Singhal et al., 2026). A large electricity instalment increase therefore reduces the resources available for other expenditures immediately and in subsequent months.

An electricity price shock combines three features that set it apart from other shocks and that underpin our argument. It is hard to avoid, it reaches each household at an idiosyncratic and clearly dated moment, and its incidence is horizontally unequal. Price increases for other goods lack the first two. A discretionary durable such as a television can be postponed or substituted, so exposure is partly a matter of choice, and a rise spread across many goods imposes a cumulative burden with no clear date for

²¹Ahlvik et al. (2026) estimate an elasticity of electricity demand with respect to its own price of approximately -0.18 in the short run during the European energy crisis. They also find that households with lower incomes have fewer opportunities to reduce consumption or increase earnings and consequently experience larger reductions in residual consumption and greater financial distress.

any one household.²² Labour market risk lacks the third. Job loss is administratively recorded, and unemployment insurance can link eligibility and benefits to the event and to prior earnings, whereas households with similar incomes can face very different electricity burdens because their consumption, dwellings, contracts, billing dates, and capacity to adjust differ [Levell et al. \(2025\)](#).²³ This horizontal inequality means that income-based redistribution is a poor instrument for compensating the households most exposed, and its political consequences remain poorly understood. Our data cannot establish them, but they warrant further study.

The second scope condition concerns the political context. Existing work shows that the radical right can gain from higher energy costs whether these stem from a deliberate government policy ([Voeten, 2024](#)) or from market prices that no government chose ([Brännlund and Peterson, 2024](#)). In each case it does so by attributing the cost to a rival's policy stance and offering a cheaper alternative. The German case is further removed from deliberate policy than either. The increase followed Russia's invasion of Ukraine, a geopolitical event outside domestic policy, while private contracts and billing schedules determined when it reached each household. The AfD nevertheless framed high prices as avoidable through weaker sanctions and renewed imports of Russian energy, and it drew voters for whom this stance was already acceptable and whose other views, especially on immigration, placed the party within reach. The instalment increase could therefore change party choice without first changing these attitudes. The government need not have caused the shock. In Germany the AfD assigned responsibility to the government's sanctions and not to any policy behind the price increase. What the cases share is a viable party that voters expected to lower their costs, could accept on other grounds,

²²Households can postpone a durable purchase, choose a cheaper product, buy a used item, or leave the market, so some avoid exposure altogether. A burden spread across many frequently purchased goods may be large in total yet less visible than a single dated bill. Such shocks may still affect political choices through perceptions of inflation or government competence, though they lack the combination central to our argument.

²³Such inequality within income groups figures prominently in welfare economics and is pervasive in the incidence of carbon pricing, where variation within income groups can exceed variation between them ([Hänsel et al., 2022](#); [Missbach and Steckel, 2026](#)). Consistent with this, an electricity price increase may raise support for redistribution, though our estimates are too imprecise to confirm or rule out this response.

and could hold responsible through a distinctive policy stance. Direct exposure benefits the radical right under these conditions. The effect should be weaker when voters regard the shock as unavoidable, expect similar future costs under all available parties, or reject the rest of the party’s platform.

6 Conclusion

In this study, we provide causal evidence that large increases in electricity costs in Germany in 2022/23 increased support for the far right populist party, *AfD*. More generally, we show that radical right parties can benefit from an economic shock even if it cannot be tied to a political decision, was not anticipated by households and got addressed by the government with large scale transfers. While our identification strategy relies on institutional features specific to the German billing system, existing evidence that radical-right parties gain from energy cost shocks in other institutional settings suggests that our findings may not be limited to the German case.

We interpret our reduced-form findings by means of a framework in which the shock raises the material advantage of the *AfD* relative to mainstream parties, changing party choice among voters who already stood close to it on the dimensions they weight most heavily – above all sanctions on Russia and continued energy imports. Attitudinal spillovers on dimensions less central to the initial switch, such as climate policy, can then entrench this realignment, so that its political consequences outlast the material shock that produced it.

Our results qualify a familiar prescription of welfare economics, which is to preserve the price signal and to compensate households for the losses it imposes – a prescription the German government followed closely when dealing with the 2022/23 energy crisis. A policy that preserves the price signal and compensates by transfer can be efficient in economic terms while proving politically costly. Those political costs, moreover, outlast the shock itself. A temporary increase in energy costs can therefore produce a durable realignment, especially if relief – albeit generous – is not closely tied to the timing of exposure to the price shock.

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A Survey questions

In this section, we report the original survey questions used in the analysis.

A.1 Socio-demographics

- *Age*: Wie alt sind Sie?
- *Female*: Welchem Geschlecht ordnen Sie sich zu?
 - Männlich; Weiblich; Divers; Keine Angabe
 - *Female* = 1 if Weiblich
- *Net monthly income of household*: Wie hoch ist das monatliche Nettoeinkommen Ihres Haushaltes insgesamt? Gemeint ist damit die Summe, die sich ergibt aus Lohn, Gehalt, Einkommen aus selbständiger Tätigkeit, Rente oder Pension, jeweils nach Abzug der Steuern und Sozialversicherungsbeiträge. Rechnen Sie bitte auch die Einkünfte aus öffentlichen Beihilfen, Einkommen aus Vermietung, Verpachtung, Wohngeld, Kindergeld und sonstige Einkünfte hinzu.
 - Unter 700 Euro; 700 bis unter 1.200 Euro; 1.200 bis unter 1.500 Euro; 1.500 bis unter 1.900 Euro; 1.900 bis unter 2.300 Euro; 2.300 bis unter 2.700 Euro; 2.700 bis unter 3.100 Euro; 3.100 bis unter 3.500 Euro; 3.500 bis unter 3.900 Euro; 3.900 bis unter 4.300 Euro; 4.300 bis unter 4.700 Euro; 4.700 bis unter 5.100 Euro; 5.100 Euro und mehr; Keine Angabe
- *Dwelling owner*: Bewohnen Sie Ihre Hauptwohnung bzw. Haus als Eigentümer?
 - Ja; Nein; weiß nicht
 - *Dwelling owner* = 1 if Ja
- *Live in eastern Germany*: In welchem Bundesland leben Sie momentan?

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- Baden-Württemberg; Bayern; Berlin; Brandenburg; Bremen; Hamburg; Hessen; Mecklenburg-Vorpommern; Niedersachsen; Nordrhein-Westfalen; Rheinland-Pfalz; Saarland; Sachsen; Sachsen-Anhalt; Schleswig-Holstein; Thüringen
 - *Live in eastern Germany* = 1 if Brandenburg or Mecklenburg-Vorpommern or Sachsen or Sachsen-Anhalt or Schleswig-Holstein or Thüringen.
- *Unemployed*: Was trifft überwiegend auf Sie zu?
 - Ich bin erwerbs- bzw. berufstätig oder selbstständig (inkl. Auszubildende, Personen in Elternzeit oder Altersteilzeit); Ich bin Schüler/-in oder Student/-in; Ich bin Rentner/-in, Pensionär/-in; Ich bin Hausfrau/-mann oder versorge Kinder und/oder pflegebedürftige Personen; Ich erhalte Sozialhilfe, Arbeitslosengeld oder Grundsicherung im Alter oder bei Erwerbsminderung; Keine der genannten Auswahlmöglichkeiten trifft auf mich zu; Keine Angabe
 - *Unemployed* = 1 if Ich erhalte Sozialhilfe, Arbeitslosengeld oder Grundsicherung im Alter oder bei Erwerbsminderung
 - *University education*: Welchen höchsten Abschluss haben Sie?
 - Keinen Abschluss (ISCED 0); Abschluss nach höchstens 8 Jahren Schulbesuch (insbesondere Abschluss im Ausland) (ISCED 1-2); Volksschul-/Realschulabschluss (Mittlere Reife), Abschluss der Polytechnischen Oberschule oder gleichwertiger Abschluss; Abitur, Fachabitur (ISCED 3-4); Lehre, Berufsausbildung im dualen System, Berufsakademie, Fachakademie (ISCED 5); Bachelorabschluss oder equivalent (ISCED 6); Masterabschluss oder equivalent (ISCED 7); Promotion oder equivalent (ISCED 8); Keine Angabe
 - *University education* = 1 if Bachelorabschluss oder equivalent (ISCED 6) or Masterabschluss oder equivalent (ISCED 7) or Promotion oder equivalent (ISCED 8)

A.2 Electricity and heating

- *Gas primary heating source*: Welcher Energieträger ist die primäre Quelle Ihrer Wärme?
 - Kohle; Öl; Holz; Gas; andere; weiß nicht
 - *Gas primary heating source* = 1 if Gas
- *Electricity from renewable*: Zahlen Sie einen Aufschlag für Ökostrom?
 - Ja; Nein; weiß nicht
 - *Electricity from renewable* = 1 if Ja

A.3 Expectations

- *Expect increase in electricity bill*: Rechnen Sie mit einer Veränderung der Abschlagszahlung für Strom nach Ihrer nächsten Jahresabrechnung?
 - Ich rechne mit einer deutlich geringeren Abschlagszahlung; Ich rechne mit einer etwas geringeren Abschlagszahlung; Ich rechne damit, dass meine Abschlagszahlung unverändert bleibt; Ich rechne mit einer etwas höheren Abschlagszahlung; Ich rechne mit einer deutlich höheren Abschlagszahlung
 - *Expect increase in electricity bill* = 1 if Ich rechne mit einer etwas höheren Abschlagszahlung or Ich rechne mit einer deutlich höheren Abschlagszahlung
- *Expect increase in heating bill*: Rechnen Sie mit einer Veränderung der Abschlagszahlung für Wärme nach Ihrer nächsten Jahresabrechnung?
 - Ich rechne mit einer deutlich geringeren Abschlagszahlung; Ich rechne mit einer etwas geringeren Abschlagszahlung; Ich rechne damit, dass meine Abschlagszahlung unverändert bleibt; Ich rechne mit einer etwas höheren Abschlagszahlung; Ich rechne mit einer deutlich höheren Abschlagszahlung
 - *Expect increase in heating bill* = 1 if Ich rechne mit einer etwas höheren Abschlagszahlung or Ich rechne mit einer deutlich höheren Abschlagszahlung

A.4 Energy behaviour

We build these variables using this question: "Bitte geben Sie an, welche der folgenden Maßnahmen Sie ergriffen haben, um die Energiepreisänderungen zu bewältigen?", with answers "Trifft voll zu"; "Trifft mit Abstrichen zu"; "Trifft eher nicht zu"; "Trifft gar nicht zu".

- *Invest in efficient appliances*: Investitionen in effizientere elektrische Geräte
 - *Invest in efficient appliances* = 1 if "Trifft voll zu" or "Trifft mit Abstrichen zu"
- *Invest in efficient heating*: Investitionen in ein effizienteres Heizsystem
 - *Invest in efficient heating* = 1 if "Trifft voll zu" or "Trifft mit Abstrichen zu"
- *Reduced car use*: Geringere Nutzung meines Kraftfahrzeugs
 - *Reduced car use* = 1 if "Trifft voll zu" or "Trifft mit Abstrichen zu"
- *Reduce temperature*: Regulierung der Raumtemperatur
 - *Reduce temperature* = 1 if "Trifft voll zu" or "Trifft mit Abstrichen zu"
- *Invest in improved insulation*: Investitionen in Wärmedämmung
 - *Invest in improved insulation* = 1 if "Trifft voll zu" or "Trifft mit Abstrichen zu"
- *Electricity savings*: Stromsparmaßnahmen
 - *Electricity savings* = 1 if "Trifft voll zu" or "Trifft mit Abstrichen zu"

A.5 Other

- *Trust Government*: Inwieweit vertrauen Sie darauf, dass die folgenden Akteure das tun, was für die Gesellschaft am besten ist?

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- überhaupt kein Vertrauen; eher geringes Vertrauen; weder geringes noch großes Vertrauen; eher großes Vertrauen; großes Vertrauen; weiß nicht
 - *Trust Government* = 1 if eher großes Vertrauen or großes Vertrauen for "Bundesregierung"
- *Climate change primarily human-induced*: Glauben Sie, dass der Klimawandel durch natürliche Prozesse, menschliches Handeln oder beide Faktoren verursacht wird?
 - nur durch natürliche Prozesse; hauptsächlich durch natürliche Prozesse; in gleichem Maße durch natürliche Prozesse und menschliches Handeln; hauptsächlich durch menschliches Handeln; nur durch menschliches Handeln; Ich glaube nicht, dass es den Klimawandel gibt; Ich weiß nicht.
 - *Climate change primarily human-induced* = 1 if hauptsächlich durch menschliches Handeln or nur durch menschliches Handeln.
- *AfD voter*: Welche politische Partei würden Sie im Falle einer Wahl derzeit am ehesten unterstützen?
 - SPD; CDU/CSU; Bündnis 90/Die Grünen; FDP; AfD; Die Linke; Eine andere Partei; Keine Partei; Weiß nicht/keine Angabe
 - *AfD voter* = 1 if AfD
- *Survey is politically neutral*: Bevor wir Ihnen die letzten Fragen stellen, würden wir gerne mehr über Ihre Wahrnehmung der Umfrage erfahren. Denken Sie, dass die Umfrage
 - politisch linkslastig ist; politisch rechtslastig ist; weder links- noch rechtslastig ist
 - *Survey is politically neutral* = 1 if weder links- noch rechtslastig ist

A.6 Indices

We provide the original questions displayed in Table 3.

- Liberal democracy: Es gibt eine Vielzahl an Meinungen zu politischen Systemen. Inwieweit stimmen Sie den folgenden Aussagen zu? (stimme überhaupt nicht zu; stimme eher nicht zu; unentschieden; stimme eher zu; stimme voll und ganz zu; weiß nicht)
 - Die Demokratie hat ihre Fehler, aber sie ist besser als jedes andere Regierungssystem.
 - Es ist am besten, eine starke politische Führung zu haben, die ein Parlament oder Wahlen nicht berücksichtigen muss und Entscheidungen schnell treffen kann.
 - Die Bundeswehr sollte ein größeres Mitspracherecht bei den Entscheidungen der Bundesregierung haben.
 - Die Bundesregierung sollte die Macht haben, politische Entscheidungen in größerem Umfang ohne Beratung mit den Landesregierungen zu treffen.
 - Politische Entscheidungen sollten verstärkt von Expertengremien getroffen werden – anstatt von gewählten Repräsentanten.
 - Das Bundesverfassungsgericht sollte weiterhin die vom Bundestag erlassenen Gesetze überprüfen und ggf. stoppen können, wenn sie mit dem Grundgesetz nicht vereinbar sind
- Redistribution: Wären Sie in den folgenden Fällen für oder gegen eine Einkommenssteuererhöhung für Besserverdienende? (stark dagegen; eher dagegen; weder dafür noch dagegen; eher dafür; stark dafür; weiß nicht)
 - die zusätzlichen Steuereinnahmen werden dafür verwendet, sozialstaatliche Hilfen für Geringverdienende auszuweiten
 - die zusätzlichen Steuereinnahmen werden dafür verwendet, mehr in die deutsche Industrie zu investieren

-
- die zusätzlichen Steuereinnahmen werden dafür verwendet, höhere Militärausgaben zu finanzieren
 - die zusätzlichen Steuereinnahmen werden dafür verwendet, den Energiekonsum zu subventionieren
 - die zusätzlichen Steuereinnahmen werden dafür verwendet, die Dekarbonisierung der Volkswirtschaft zu finanzieren
 - die zusätzlichen Steuereinnahmen werden dafür verwendet, andere Steuern zu reduzieren
- Climate policies: Inwieweit Sind Sie für oder gegen die folgenden Klimaschutzmaßnahmen? (stark dagegen; eher dagegen; weder dafür noch dagegen; eher dafür; stark dafür; weiß nicht)
 - Einheitliche, also alle Sektoren umfassende, CO₂-Steuer, um die Preise für alle fossilen Brennstoffe zu erhöhen
 - Subventionierung des Ausbaus erneuerbarer Energie, wie z.B. Wind und Solar
 - Laufzeitverlängerung der Atomkraftwerke
 - Wiedereinstieg in die Atomenergie
 - Subventionierung von Investitionen in die verbesserte Wärmedämmung von Häusern, um den beim Heizen bzw. Kühlen entstehenden Energiebedarf zu reduzieren
 - Finanzielle Hilfe für Schwellen- und Entwicklungsländer zur Bekämpfung des Klimawandels
 - Xenophobia: Es gibt unterschiedliche Meinungen zur Rolle von Immigranten und Minderheiten in Deutschland. Beachten Sie, dass es keine richtigen oder falschen Antworten gibt, da Menschen unterschiedliche Erfahrungen gemacht haben. Inwieweit stimmen Sie den folgenden Aussagen zu? (stimme überhaupt nicht zu; stimme eher nicht zu; unentschieden; stimme eher zu; stimme stark zu; weiß nicht)

-
- Deutschland sollte eine multikulturelle Nation sein.
 - Immigranten sollten sich an deutsche Traditionen bzw. an die deutsche Lebensweise anpassen.
 - Immigranten sollten in gleichem Umfang wie alle anderen deutschen Staatsbürger:innen Zugang zu sozialstaatlichen Hilfen haben.
 - Die Bundesregierung sollte bei der Verwendung von öffentlichen Unterstützungsleistungen deutsche Staatsbürger:innen bevorzugt behandeln.
 - Deutschland sollte diejenigen, die vor Krieg oder politischer Unterdrückung fliehen, aufnehmen.
 - Kriminelle mit ausländischer Staatsangehörigkeit sollten systematisch abgeschoben werden.
 - Die deutsche Staatsangehörigkeit sollte allen in Deutschland Geborenen ohne Vorbedingungen zugestanden werden. (Geburtsortsprinzip).
 - Deutschland kann viele Immigranten aufnehmen, ohne die innere Einheit zu gefährden.
 - Es scheint so, als wollten Minderheiten nicht gleiche Rechte - sondern Sonderrechte.

A.7 Treatment variables

For the binary treatment, we ask the following question: "Hat sich die Höhe Ihrer Abschlagszahlung für Strom seit der letzten Befragung verändert?" ("Has the amount of your advance payment for electricity changed since the last survey?"), with possible answers: "Ja, meine Abschlagszahlung hat sich erhöht"; "Ja, meine Abschlagszahlung hat sich verringert", "Nein" "weiß nicht". We classify the individuals that have received an increase: "Ja, meine Abschlagszahlung hat sich erhöht", as treated.

The definition of the continuous treatment is slightly more complicated. During the first wave, we ask respondent to provide their current instalment payments for electricity: "Bitte geben Sie die Höhe Ihrer monatlichen Abschlagszahlung [für Strom] an.". Later,

we ask them to report the change in instalment payments if they are treated with the binary treatment: "Bitte geben Sie an, um wieviel sich Ihre monatliche Abschlagszahlung verändert hat, d.h. wieviel mehr oder weniger Sie für Ihre monatliche Stromnutzung zahlen?". We then build the continuous treatment by combining the baseline payment with the reported variation.

B Supplementary tables

B.1 Treatment timing

Table 13: Pre-treatment socio-economic characteristics: Early-treated vs. never-treated

	Individuals treated in wave 2
Female	-0.041 (0.028)
Age	0.002** (0.001)
Net monthly income of household (€)	0.000 (0.007)
Unemployed	0.071 (0.052)
University education	0.015 (0.033)
Live in eastern Germany	-0.029 (0.038)
Dwelling surface	0.000 (0.000)
House owner	0.042 (0.034)
R-squared	0.014
Observations	795
<i>Note:</i> Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

Table 14: Pre-treatment political attitudes: Early-treated vs. never-treated

	Individuals treated in wave 2
Female	-0.041 (0.028)
Age	0.002** (0.001)
Net monthly income of household (€)	0.000 (0.007)
Unemployed	0.071 (0.052)
University education	0.015 (0.033)
Live in eastern Germany	-0.029 (0.038)
Dwelling surface	0.000 (0.000)
House owner	0.042 (0.034)
R-squared	0.014
Observations	795
<i>Note:</i> Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

Table 15: Pre-treatment partisan affiliation: Early-treated vs. never-treated

	Individuals treated in wave 2
Female	-0.041 (0.028)
Age	0.002** (0.001)
Net monthly income of household (€)	0.000 (0.007)
Unemployed	0.071 (0.052)
University education	0.015 (0.033)
Live in eastern Germany	-0.029 (0.038)
Dwelling surface	0.000 (0.000)
House owner	0.042 (0.034)
R-squared	0.014
Observations	795

Note: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Early-treated vs. never-treated

Table 16: Treatment timing and pre-treatment socio-economic characteristics

	Individuals treated in wave 2
Female	-0.126** (0.064)
Age	-0.003 (0.002)
Net monthly income of household (€)	0.005 (0.014)
Unemployed	0.176 (0.110)
University education	0.055 (0.071)
Live in eastern Germany	-0.068 (0.085)
Dwelling surface	0.000 (0.001)
House owner	0.113 (0.076)
R-squared	0.046
Observations	257

Note: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 17: Treatment timing and pre-treatment political attitudes

	Individuals treated in wave 2
Female	0.009 (0.025)
Age	0.001 (0.001)
Net monthly income of household (€)	-0.005 (0.006)
Unemployed	0.059 (0.046)
University education	-0.039 (0.030)
Live in eastern Germany	-0.015 (0.034)
Dwelling surface	0.000* (0.000)
House owner	0.070** (0.030)
R-squared	0.020
Observations	656
<i>Note:</i> Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

Table 18: Treatment timing and pre-treatment partisan affiliation

	Individuals treated in wave 2
Female	0.009 (0.025)
Age	0.001 (0.001)
Net monthly income of household (€)	-0.005 (0.006)
Unemployed	0.059 (0.046)
University education	-0.039 (0.030)
Live in eastern Germany	-0.015 (0.034)
Dwelling surface	0.000* (0.000)
House owner	0.070** (0.030)
R-squared	0.020
Observations	656
<i>Note:</i> Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$	

Early-treated vs. late-treated for an above median increase in installment payments

B.2 No anticipation

Early-treated vs. never-treated

Table 19: Pre-test of no anticipation on partisan affiliation: Early-treated vs. never-treated

Partisan Affiliation	Estimate (S.E)	p value	Observations
RWPs support (AfD)	0.009 (0.019)	0.634	3560
Social-democrats support (SPD)	0.044 (0.022)	0.045	3560
Conservatives support (CDU)	0.026 (0.024)	0.277	3560
Greens support (Grünen)	-0.019 (0.021)	0.355	3560
Liberals support (FDP)	-0.018 (0.013)	0.155	3560
Radical left support (Die Linke)	-0.001 (0.012)	0.923	3560
No party affiliation	-0.037 (0.024)	0.119	3560

Table 20: Pre-test of no anticipation on political attitudes: Early-treated vs. never-treated

Political attitudes	Estimate (S.E)	p value	Observations
Index trust in political institutions	-0.078 (0.062)	0.208	3345
Index Populism	-0.037 (0.060)	0.535	3316
Index pro redistribution	0.017 (0.063)	0.787	3187
Index pro climate policies	-0.06 (0.061)	0.300	3233
Index xenophobia	0.010 (0.061)	0.867	3315

Table 21: Pre-test of no anticipation on partisan affiliation

Political Affiliation	Estimate (S.E)	p value	Observations
RWPs support (AfD)	-0.013 (0.057)	0.815	207
Social-democrats support (SPD)	0.130 (0.053)	0.016	207
Conservatives support (CDU)	-0.030 (0.062)	0.629	207
Greens support (Grünen)	0.029 (0.052)	0.583	207
Liberals support (FDP)	0.043 (0.028)	0.121	207
Radical left support (Die Linke)	-0.082 (0.047)	0.084	207
No party affiliation	-0.128 (0.077)	0.098	207

Early-treated vs. late-treated for an above median increase in installment payments

Table 22: Pre-test of no anticipation on political attitudes

Political Attitudes	Estimate (S.E)	p value	Observations
Index trust in political institutions	0.085 (0.161)	0.597	205
Index Populism	0.044 (0.164)	0.790	193
Index pro redistribution	0.116 (0.165)	0.483	186
Index pro climate policies	0.218 (0.171)	0.204	190
Index xenophobia	-0.131 (0.149)	0.381	198

B.3 Parallel trends

Table 23: Pre-test of parallel trends for individuals treated at $t = 3$ vs. never-treated

Political Attitudes	Estimate (S.E)	p value	Observations
Index trust in political institutions	0.065 (0.088)	0.457	1210
Index Populism	-0.194 (0.100)	0.054	1188
Index pro redistribution	0.132 (0.161)	0.415	1142
Index pro climate policies	0.001 (0.071)	0.992	1170
Index xenophobia	0.012 (0.089)	0.897	1200

Table 24: Pre-test of parallel trends for individuals treated at $t = 3$ vs. never-treated

Partisan Affiliation	Estimate (S.E)	p value	Observations
RWPs support (AfD)	-0.031 (0.017)	0.073	1296
Social-democrats support (SPD)	0.003 (0.036)	0.925	1296
Conservatives support (CDU)	0.039 (0.024)	0.098	1296
Greens support (Grünen)	-0.044 (0.046)	0.336	1296
Liberals support (FDP)	0.010 (0.006)	0.109	1296
Radical left support (Die Linke)	-0.022 (0.017)	0.180	1296
No party affiliation	0.046 (0.042)	0.274	1296

Early-treated vs. never-treated

Table 25: Pre-test of parallel trends for individuals treated at $t = 3$ vs. $t = 4$ for an above median increase in installment payments

Political Attitudes	Estimate (S.E)	p value	Observations
Index trust in political institutions	-0.148 (0.250)	0.556	82
Index Populism	-0.099 (0.255)	0.699	74
Index pro redistribution	0.343 (0.432)	0.433	68
Index pro climate policies	0.130 (0.224)	0.565	72
Index xenophobia	-0.005 (0.296)	0.986	78

Table 26: Pre-test of parallel trends for individuals treated at $t = 3$ vs. $t = 4$ for an above median increase in installment payments

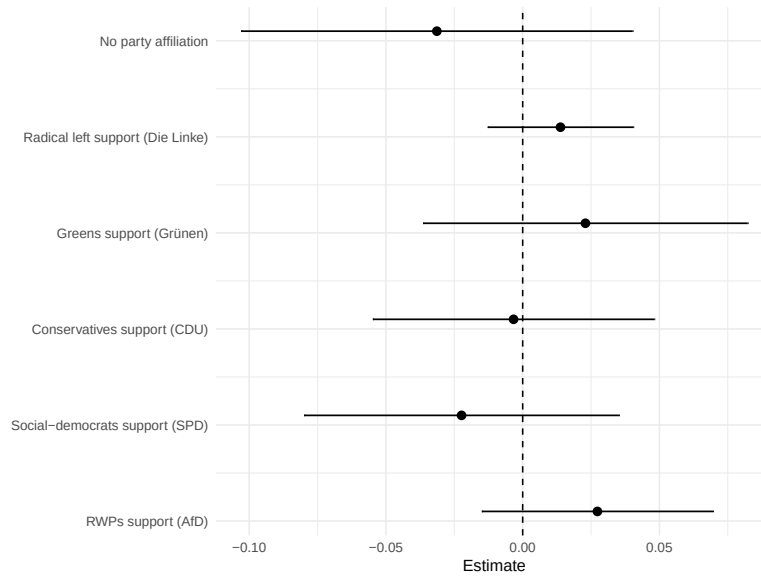
Partisan Affiliation	Estimate (S.E)	p value	Observations
RWPs support (AfD)	-0.037 (0.037)	0.326	82
Social-democrats support (SPD)	0.074 (0.052)	0.159	82
Conservatives support (CDU)	0.037 (0.037)	0.326	82
No party affiliation	0.037 (0.085)	0.664	82

Early-treated vs. late-treated for an above median increase in installment payments

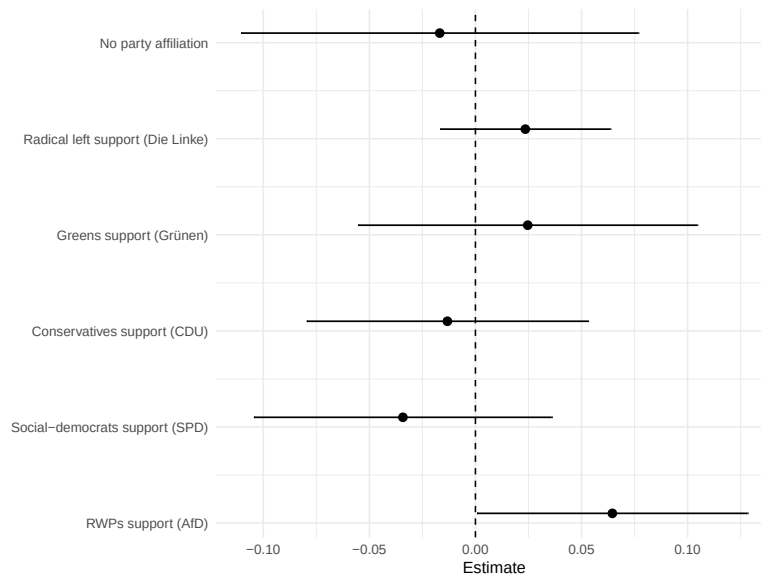
C Supplementary Figures

C.1 Main results with covariates

Figure 6: Conditional D-i-D effects of receiving a higher electricity bill on partisan affiliation.

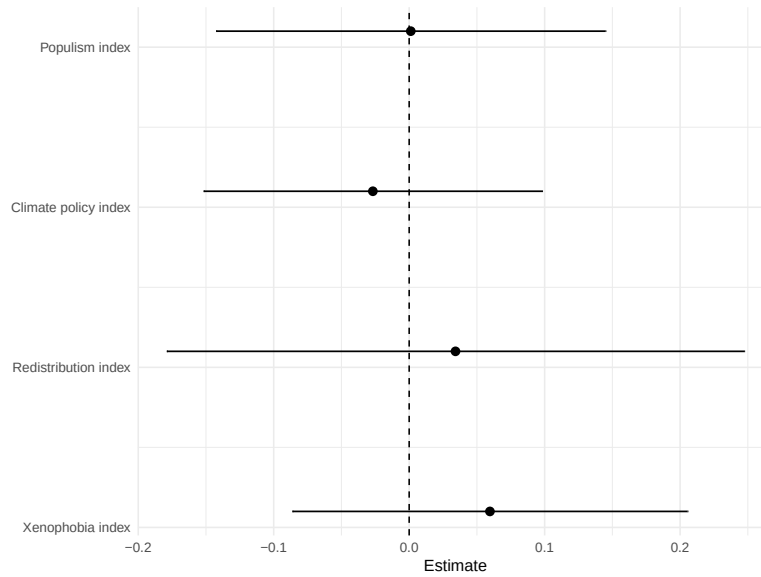


(a) ATT of the binary treatment: 'Increased electricity costs'

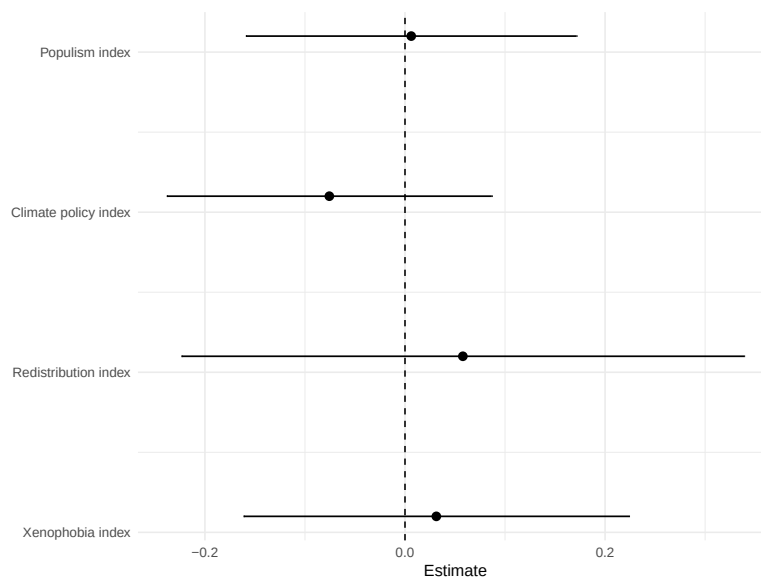


(b) ATT of the binary treatment: 'Above median increase in electricity costs'

Figure 7: Conditional D-i-D effects of receiving a higher electricity bill on political attitudes.



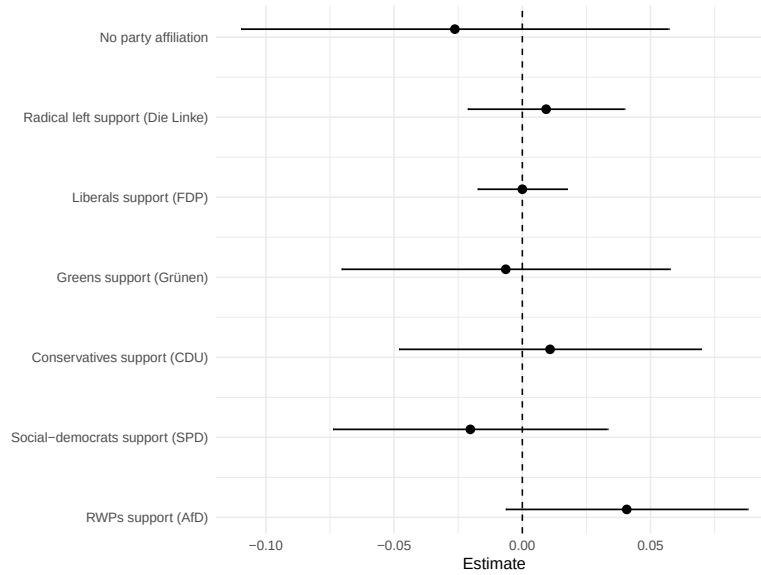
(a) ATT of the binary treatment: 'Increased electricity costs'



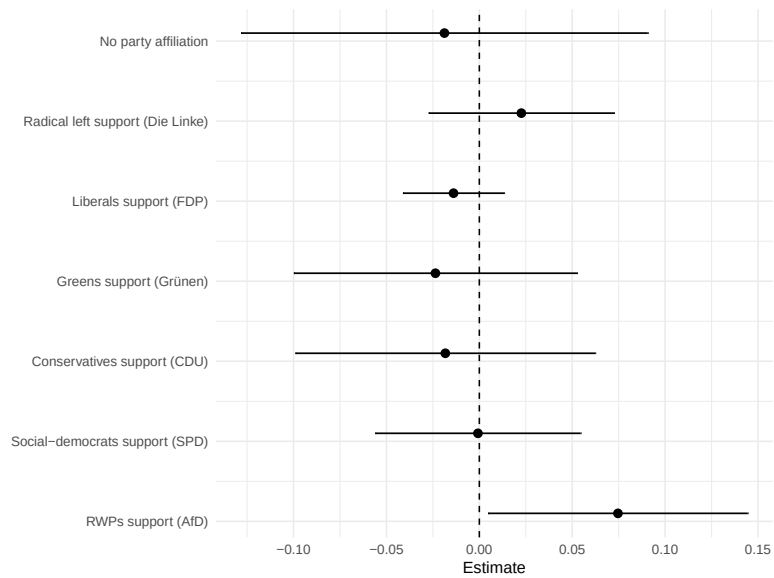
(b) ATT of the binary treatment: 'Above median increase in electricity costs'

C.2 Main results for the balanced panel data

Figure 8: Conditional D-i-D effects of receiving a higher electricity bill on partisan affiliation.

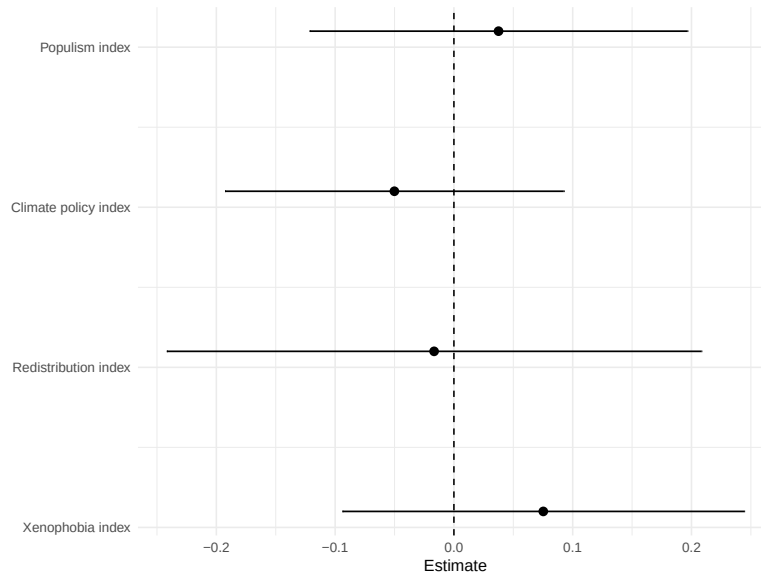


(a) ATT of the binary treatment: 'Increased electricity costs'

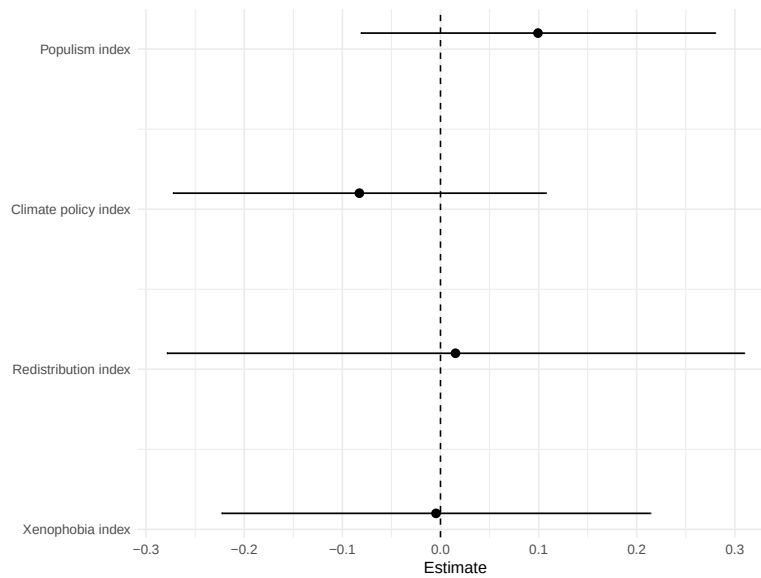


(b) ATT of the binary treatment: 'Above median increase in electricity costs'

Figure 9: Unconditional D-i-D effects of receiving a higher electricity bill on political attitudes.



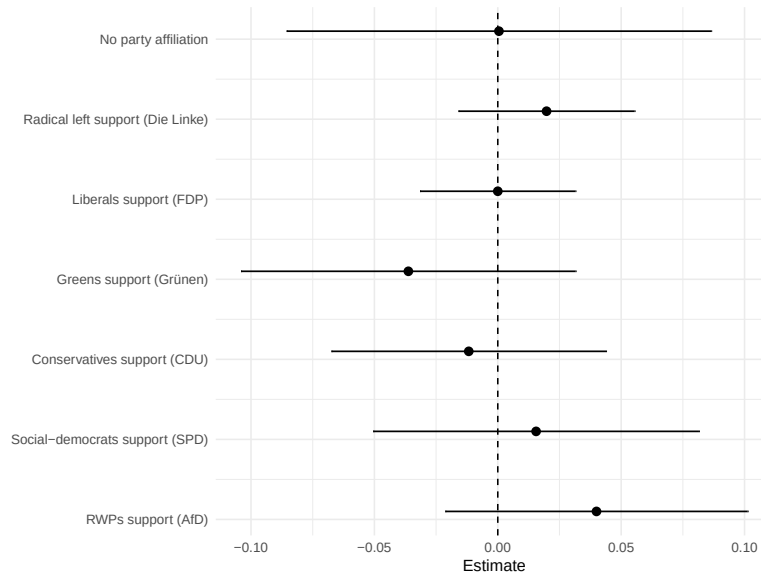
(a) ATT of the binary treatment: 'Increased electricity costs'



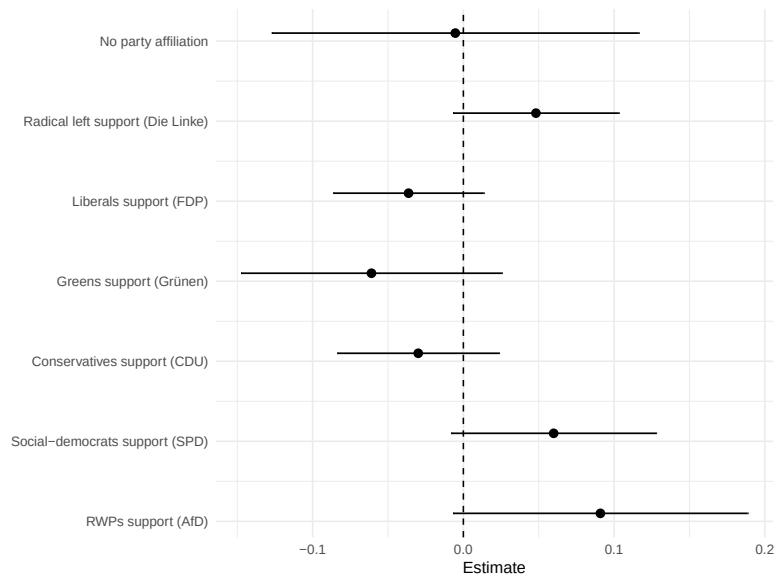
(b) ATT of the binary treatment: 'Above median increase in electricity costs'

C.3 Main results for individuals treated in wave 2 (May-June 2023)

Figure 10: Unconditional D-i-D effects of receiving a higher electricity bill on partisan affiliation.

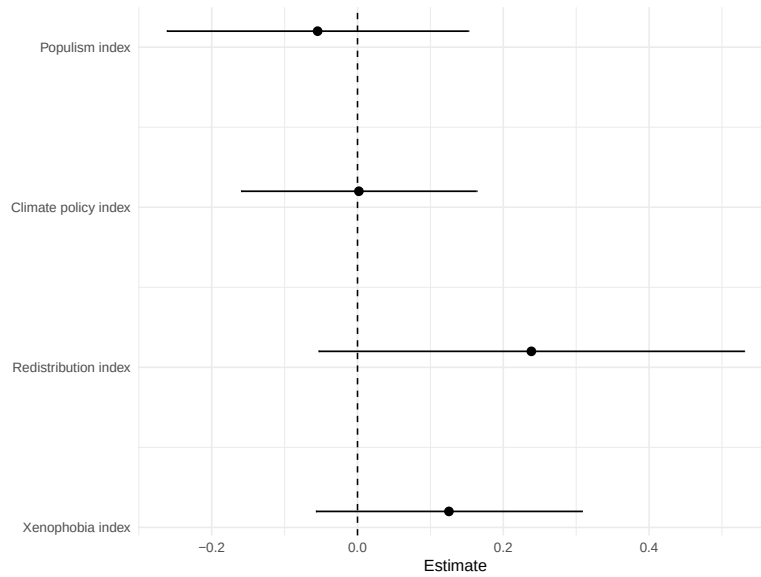


(a) ATT of the binary treatment: 'Increased electricity costs'

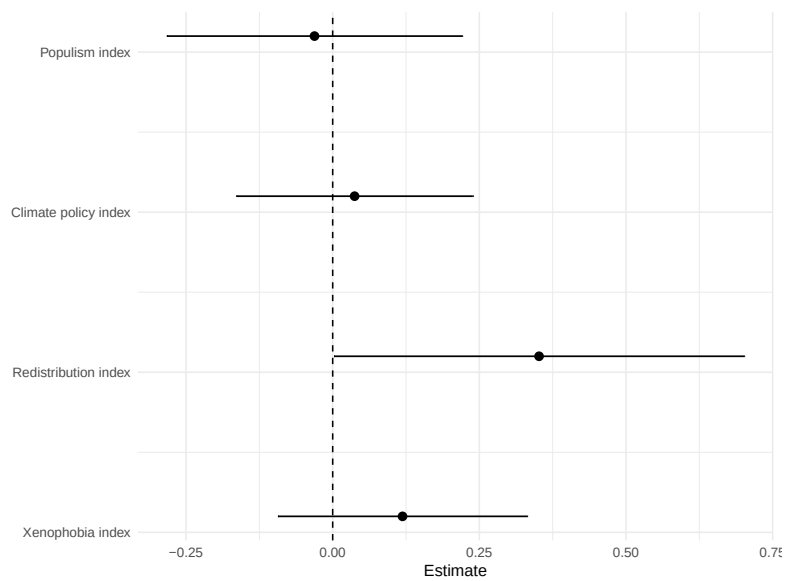


(b) ATT of the binary treatment: 'Above median increase in electricity costs'

Figure 11: Unconditional D-i-D effects of receiving a higher electricity bill on political attitudes.



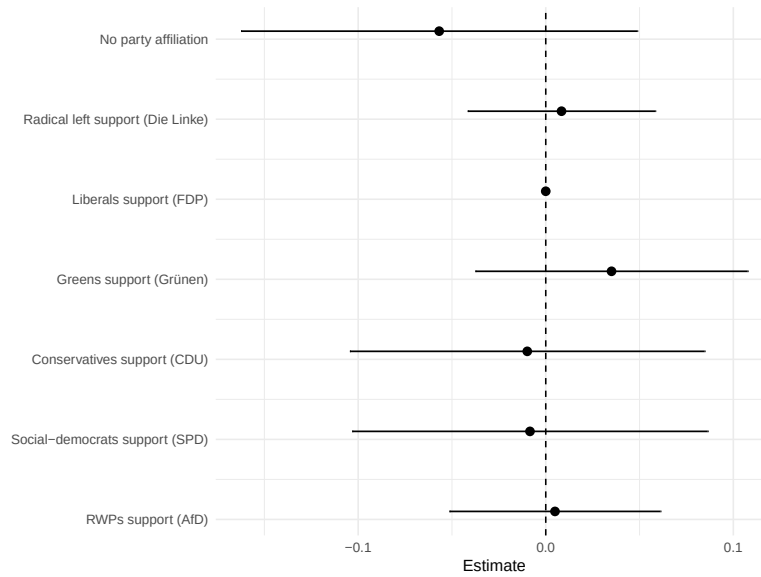
(a) ATT of the binary treatment: 'Increased electricity costs'



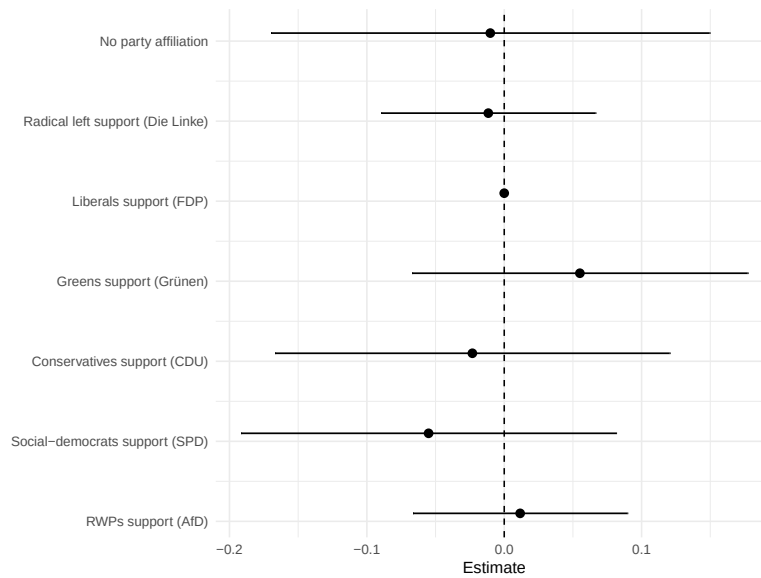
(b) ATT of the binary treatment: 'Above median increase in electricity costs'

C.4 Main results for individuals treated in wave 3 (October-November 2023)

Figure 12: Unconditional D-i-D effects of receiving a higher electricity bill on partisan affiliation.

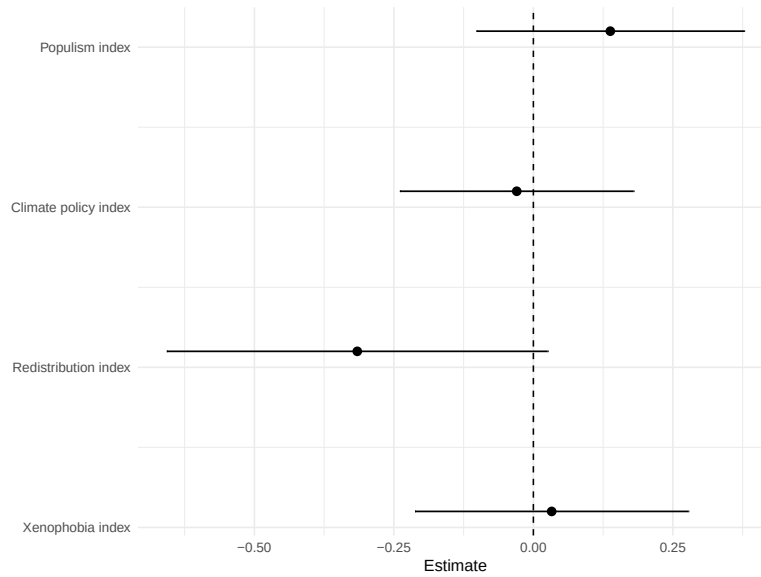


(a) ATT of the binary treatment: 'Increased electricity costs'

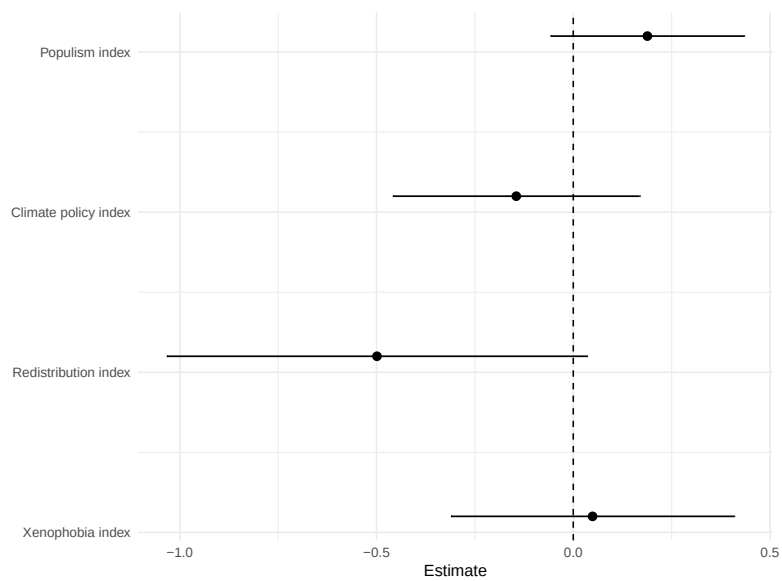


(b) ATT of the binary treatment: 'Above median increase in electricity costs'

Figure 13: Unconditional D-i-D effects of receiving a higher electricity bill on political attitudes.



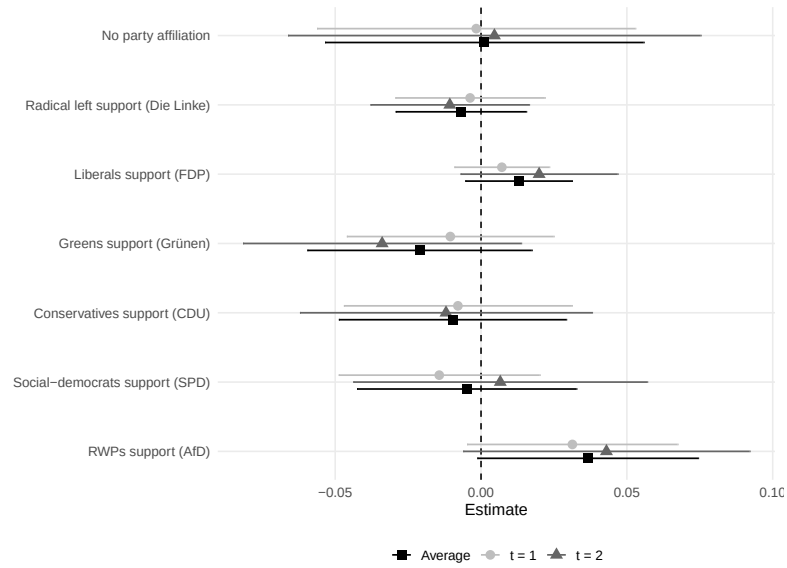
(a) ATT of the binary treatment: 'Increased electricity costs'



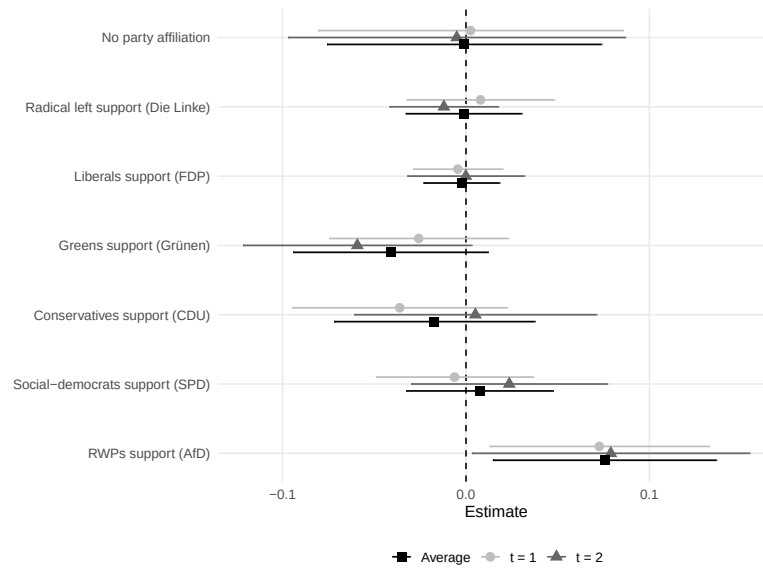
(b) ATT of the binary treatment: 'Above median increase in electricity costs'

C.5 Main results using never-treated individuals as control group in the balanced panel

Figure 14: Unconditional D-i-D effects of receiving a higher electricity bill on partisan affiliation.

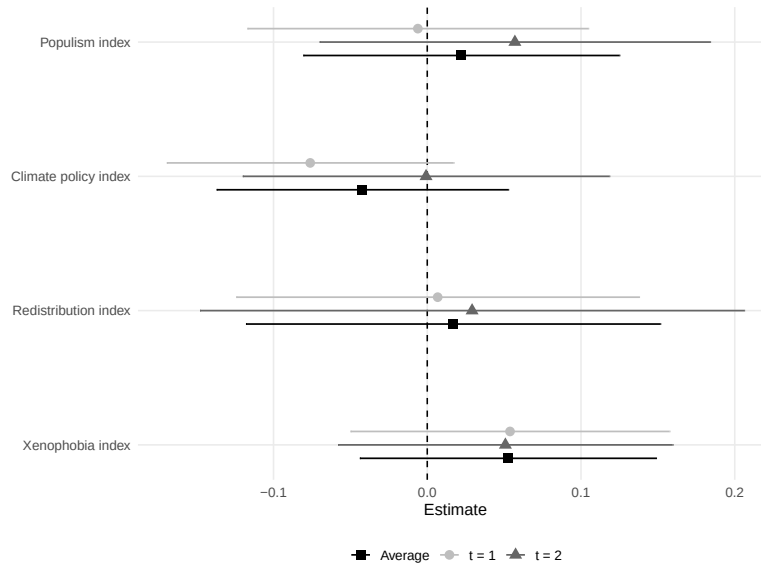


(a) ATT of the binary treatment: 'Increased electricity costs'

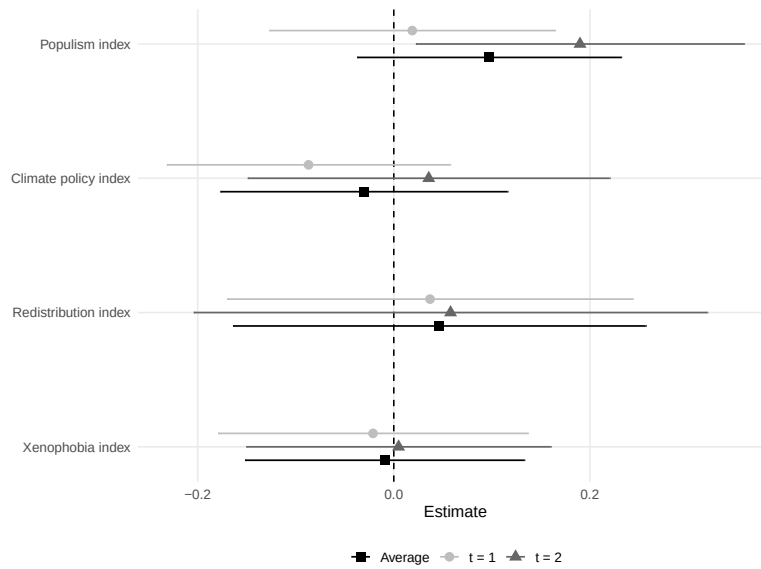


(b) ATT of the binary treatment: 'Above median increase in electricity costs'

Figure 15: Unconditional D-i-D effects of receiving a higher electricity bill on political attitudes.



(a) ATT of the binary treatment: 'Increased electricity costs'



(b) ATT of the binary treatment: 'Above median increase in electricity costs'

D Supplementary exploratory analyses

Our results suggest that large electricity price increases trigger people to support the right wing populist party AfD. In the following, we will discuss (i) the role of government compensation in political shifts, (ii) whether shifts in voting intention are likely persistent, and (iii) which additional theoretical mechanisms can explain those shifts.

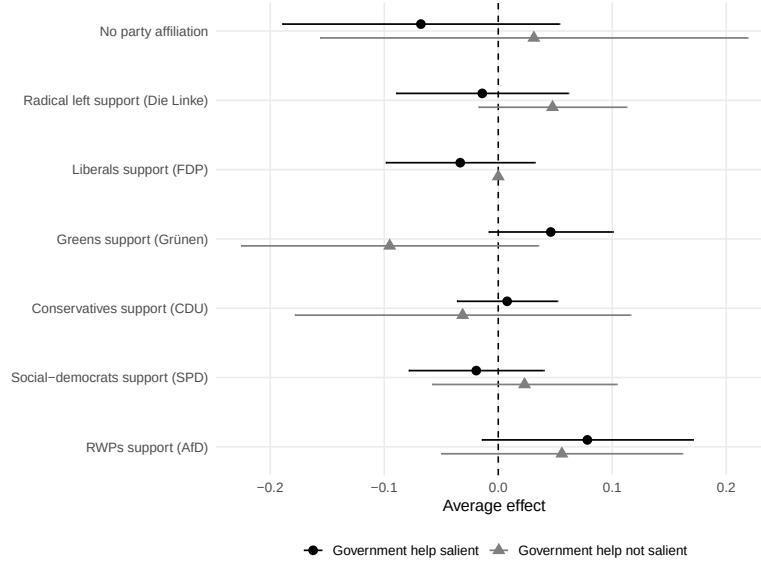
D.1 Salience of compensation

While our main analysis focuses on the effects of electricity price increases, it is informative to examine whether compensatory government transfers moderated political responses. Before and during the period under study, most (if not all) households in Germany received compensation aimed at offsetting rising electricity costs (for details see section 2). However, survey evidence reveals substantial heterogeneity in whether households understood that they had been compensated.

We exploit this variation by estimating a difference-in-differences model in which treatment is defined as the salience of compensation, i.e., whether respondents report understanding that they received compensation for higher energy-related costs in the first wave of the panel.

Figure 16 shows no statistically significant difference in support for the AfD following the receipt of increased electricity payments depending on salience.

Figure 16: Unconditional D-i-D effects of receiving an above median increase in costs bill on partisan support, among respondents who were/were not aware of governmental measures to dampen the effect of the energy crisis, measured in January 2023.



Compensation, even when salient, did not affect shifts in political attitudes, which indicate that objective income transfers may have limited political effects.

D.2 Persistence of voting intention shifts

One important question to discuss is whether the initial impetus to vote for AfD translates into sustained support of the right wing populist party. We find evidence for dynamics effects of the electricity shock. Figure 14 in Appendix shows that voters are likely to keep supporting the AfD several months after the initial initial increase.²⁴

In this regard, the divergence between a persistent shift in voting intention without a significant effect on underlying attitudes might seem surprising. Yet, this pattern is not without precedent. Voting intentions, being relatively low costs and affective in nature, can shift quickly in response to a salient shock, while underlying attitudes adjust more slowly (Norris and Inglehart, 2019). Consistent with this view, we find that

²⁴We use never-treated units as the control group when measuring dynamic effects because of sample size limitations with not-yet-treated individuals.

populist attitudes increase only in the second wave after the initial treatment (Figure 15, in Appendix). Transforming material hardship into a structured populist disposition plausibly requires time, which would account for the observed delay.

Lastly, we are interested in the effect that supporting the AfD might have on political attitudes. [Callander and Carbajal \(2022\)](#) hypothesise that changes in political attitudes is partly driven by citizens reducing the ideological gap to their favourite party, and partisan elites who strategically radicalise their voters. If this hypothesis is correct, we would expect that respondents who switch to supporting the AfD would also converge over time towards the policy positions defending by the party. We provide some additional evidence on the policy attitudes of individuals who switched voting intentions to the right-wing populist party during our data collection. In particular, we compare the trajectories of respondents who started to support the AfD in the third wave of our panel to (i) people who never expressed support for the AfD and (ii) people we support the AfD in all 4 waves.²⁵ Figure 17 shows that respondents who eventually switch to the AfD have initial attitudes similar to those who are 'long-term' AfD supporters, in particular when it comes to xenophobic attitudes and support for redistribution. We also observe a drop in support for climate policies in wave 4, a few months after respondents picked the AfD as their favourite party. We further estimate the following fixed effects model to test the potential change in political attitudes of 'new' AfD voters compared to 'long-term' AfD supporters:

$$Y_{it} = \beta_1 (1\{t = 1\} \times \text{New}_i) + \beta_3 (1\{t = 3\} \times \text{New}_i) + \beta_4 (1\{t = 4\} \times \text{New}_i) + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (10)$$

where new AfD voters follow this trajectory

$$\text{New}_i = 1 \Leftrightarrow \{AfD_{i1} = 0, AfD_{i2} = 0, AfD_{i3} = 1, AfD_{i4} = 1\},$$

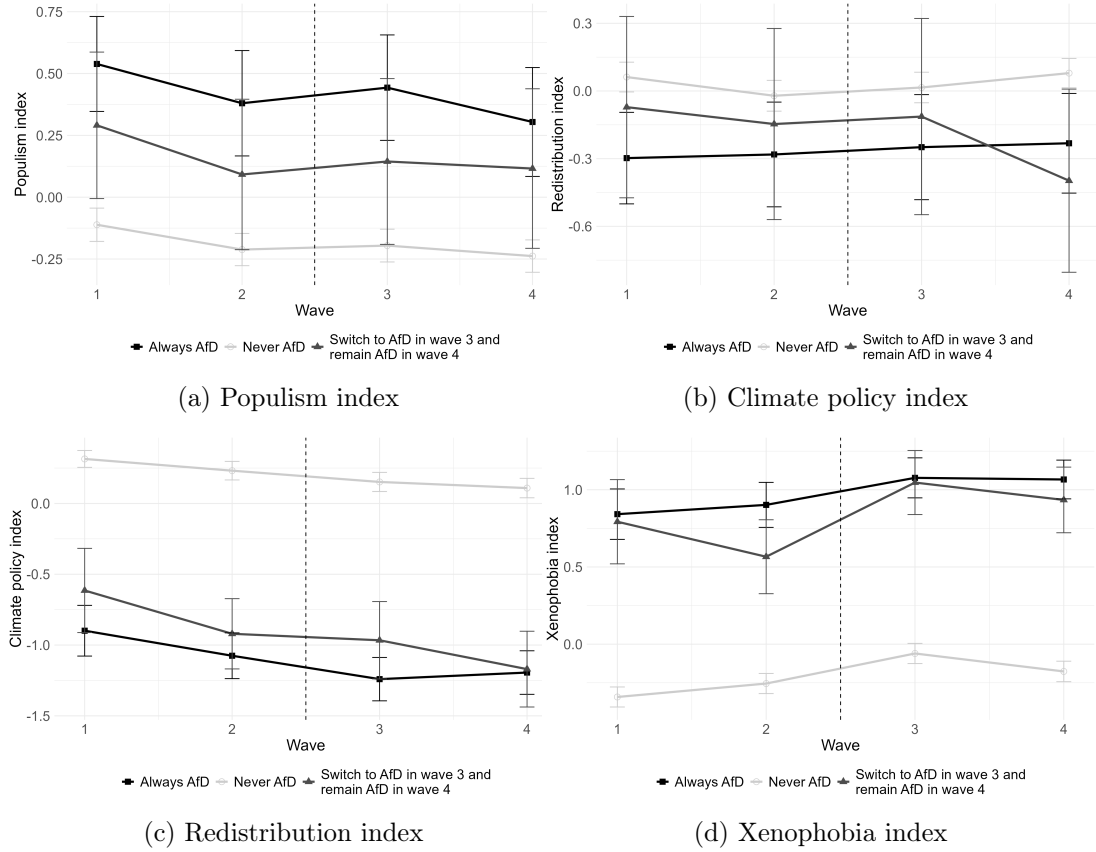
and the control group consists exclusively of respondent who always support the AfD

²⁵We use switchers in the third wave in order to have an estimate of the trends in attitudes before the switch to the AfD.

during the panel

$$\text{New}_i = 0 \Leftrightarrow \{AfD_{it} = 1 \ \forall t\}.$$

Figure 17: Political attitudes of respondents who (i) never express the intention to vote for the AfD, (ii) always express the intention to vote for the AfD, and (iii) have switched voting intentions in wave 3.



We find that attitudes of new AfD voters do not become significantly more similar to 'long-term' AfD voters compared to the pre-switch period. However, we find that individuals become significantly more opposed to redistributive and climate policies between waves 3 and 4, a few months after they start supporting the AfD.

While those results cannot be interpreted in a causal way, they suggest that individuals that have shifted voting intention towards populists will not easily shift back towards more liberal parties. This could lead to a *populist trap* (Callander and Carbajal, 2022;

	Xenophobia index	Redistribution index	Climate policy index	Populism index	in-
Wave 1 $\times$ New	0.287** (0.139)	0.091 (0.176)	0.129 (0.139)	0.040 (0.131)	
Wave 3 $\times$ New	0.305** (0.128)	0.001 (0.180)	0.119 (0.107)	-0.011 (0.161)	
Wave 4 $\times$ New	0.203 (0.143)	-0.301 (0.186)	-0.131 (0.104)	0.100 (0.144)	
Observations	508	488	504	500	
FE: ID + Wave	Yes	Yes	Yes	Yes	
Wave 4 – Wave 3	-0.102	-0.302	-0.250	0.110	
p -value (H_0 : Wave 4 = Wave 3)	0.352	0.065	0.006	0.478	

Notes: Robust standard errors clustered at the respondent level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 27: Difference over time in political attitudes between individuals who started to support the AfD between waves 2 and 3 (*New*) and individuals who support the AfD in all 4 waves.

[Herrera et al., 2024](#)), where people start voting for populist parties because of economic shocks and continue to do so after the shock is gone because supporting populist parties has changed their world-views and political preferences.

Understanding why the shift in voting intention appears durable requires unpacking the underlying channels through which electricity prices increases translate into political realignment, which we will turn to in the following section.

D.3 Additional mechanisms

Building on the political economy of populism, we discuss four theories that could explain our reduced-form effect of higher energy prices increasing the probability of supporting right-wing populist parties: the role of redistribution, energy poverty, welfare chauvinism and attributional learning. Each of these theories implies different causal mechanisms. In the following, we focus on explicating these mechanisms and present suggestive evi-

dence.²⁶

Right-wing populism and redistribution An increase in energy prices reduces household's disposable income. Following the Meltzer-Richard logic ([Meltzer and Richard, 1981](#)), this should increase demand for redistribution for self-interested reasons.²⁷ Usually, the argument is that redistributive preferences should translate into a higher probability of supporting 'redistributive' parties, those parties that run on redistributive platforms, which are typically those on the left ([Rueda and Stegmüller, 2019](#)). For redistributive concerns to result in right-wing populist support, we thus have to consider alternative channels that explain why voters not only turn away from the government, but turn to the AfD, rather than left-wing opposition parties (notably the radical left party 'Die Linke'). While punishing the government can be explained by, for instance, people believing the latter has not compensated them sufficiently for the energy price shock, this type of sanctioning logic, centred on redistributive concerns, does not explain the growing appeal of right-wing populists.

We look at two alternative channels. First, we consider the level differences between respondents in their beliefs about the importance of offering costly support to Ukraine as a central moderating variables in our context. Consequently, people who are unwilling to incur substantial cost for supporting Ukraine in the form of, e.g., tax revenues used for funding military support or higher electricity prices entailed by embargoing Russian oil and gas. Indeed, the AfD was the only party that consistently and clearly argued for repealing the embargo on Russia oil and gas, which was introduced to sanction Russia for waging war on Ukraine, so as to ensure the supply of affordable energy. In this sense, supporting the AfD could have been seen as redistributive in our context, despite the

²⁶The split sample regressions we show rely on small sample sizes, and should be interpreted with caution.

²⁷In the Meltzer-Richard model, voters choose their preferred level of redistribution based on how taxation and transfers affect their disposable income. Because taxes are linear and revenues are redistributed equally, voters below mean income prefer positive redistribution, while those at or above mean income prefer little or none. With single-peaked preferences and one-dimensional party competition, parties converge to the tax rate preferred by the median voter. The model therefore predicts that as inequality rises and the gap between mean and median income widens, political demand for redistribution—and hence equilibrium redistribution—also increases

lack of increased support for traditional tax-based redistributive policies (captured in our index 'Support for redistribution').

Second, for those adversely affected by the energy price shock, turning to the AfD is more appealing than supporting parties that promise more conventional tax-based redistribution, only if they believe that government funds are used efficiently (Culpepper et al., 2025). The central reason is that believing the government to be wasteful means that promises of compensation or redistribution will lack credibility.

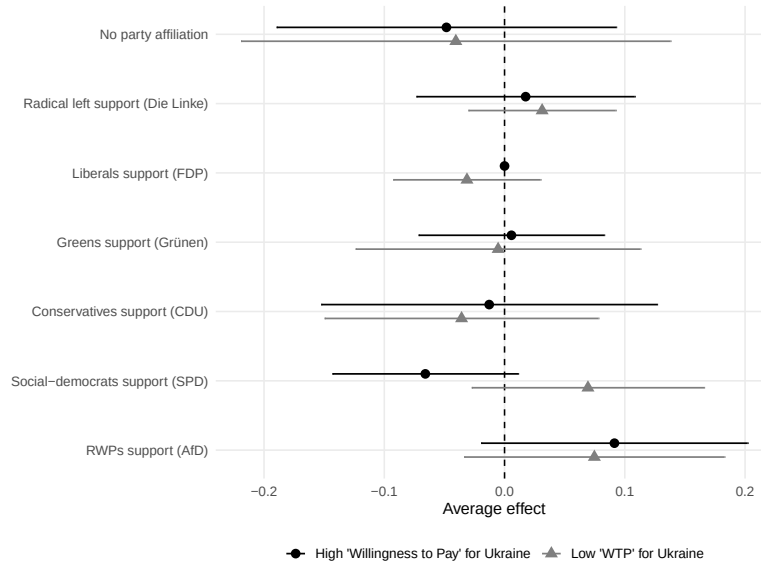
We aim at providing suggestive evidence on both channels using supplementary survey questions. We measure the willingness to pay for Ukraine by computing the z-score of answers to two survey questions: 'Germany should support Ukraine financially and provide humanitarian aid.' and 'Germany should continue to import Russian gas and oil to ensure the supply of affordable energy.'²⁸ We capture perception of government efficiency via a survey item asking people to estimate the cent amount that the government wastes for each euro of tax revenue. We capture the saliency of compensation efforts with a survey item asking people if they received any governmental help for their energy expenses.²⁹ Then we calculate the effect of increasing electricity costs among different sub-samples: respondents with above and below average willingness to pay for Ukraine in wave 1 (pre-treatment), and respondents with above and below average perception of government waste in wave 1.

Our split sample analysis does not show evidence for either channels (Figure 18). We find that the increase in support for the AfD is not particularly strong among respondents who, pre-treatment, were less willing to bear financial costs to support Ukraine. Similarly, AfD support rises by a similar magnitude irrespective of perceptions of governmental waste.

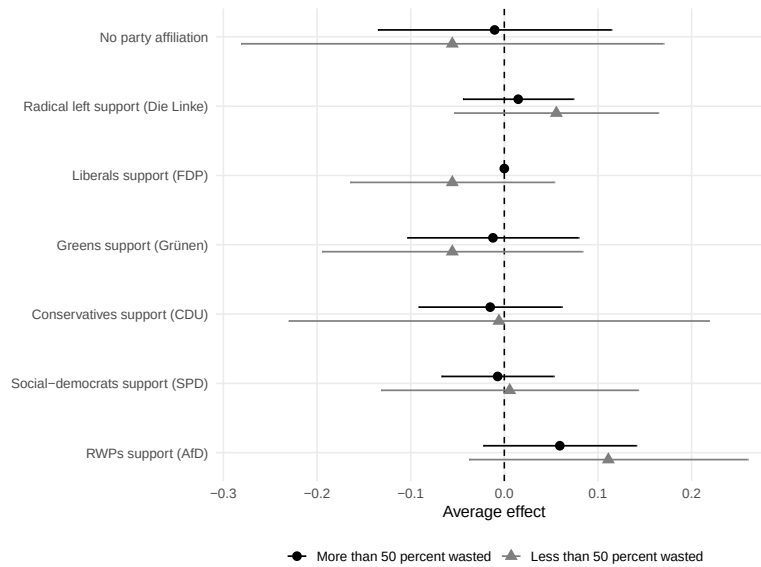
²⁸The first item contributes positively to the z-score and the second negatively.

²⁹In theory, every respondent should have received some form of governmental help, see Section 2

Figure 18: Unconditional D-i-D effects of receiving a higher electricity bill on partisan support: Split sample on welfare and income characteristics



(a) ATT of the binary treatment: 'Above median increase in electricity costs' among respondents with below/above average willingness to pay for Ukraine

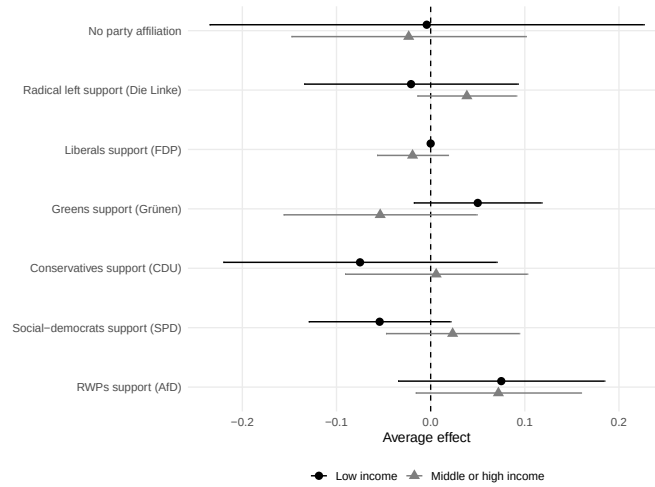


(b) ATT of the binary treatment: 'Above median increase in electricity costs' among respondents with perceptions of high/low governmental waste

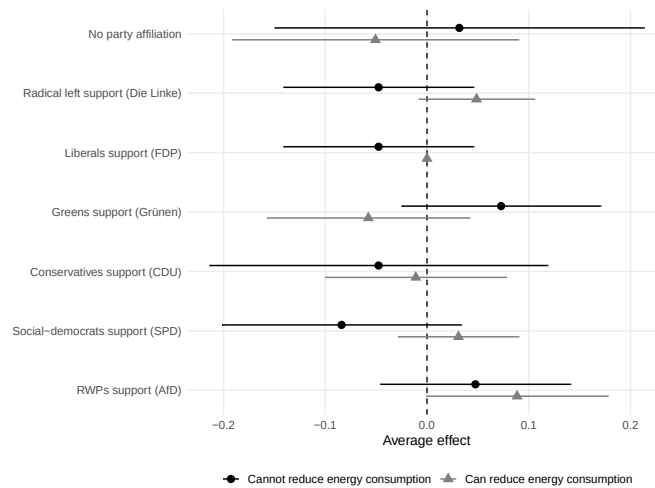
Energy consumption and energy poverty The second mechanism we consider is that the energy price shock incentivises households to cut back on their energy consumption. Those unable to do so because their energy consumption is just sufficient to cover basic needs might suffer from high welfare losses, which could lead to an increase of perceived unfairness due to energy price shocks and potentially increasing AfD support (Guriev and Papaioannou, 2022, make a similar point, noting that support for populists can originate from perceptions of unfairness). In particular, lower-income households also tend to spend a larger share of their total expenditures on energy consumption, increasing their relative exposure to energy price shocks. Similarly, people with lower-quality housing insulation or harsher climates experience greater difficulty in reducing electricity and heating consumption, which could lead to resentment towards political incumbents.

Relying on split sample analysis, we do not find empirical patterns consistent with this mechanism (Figure 19). Compared to middle/high income households, low-income households are not more likely to support the AfD after receiving a higher electricity bill, it is not the case for people who state that they cannot or have difficulties to reduce their energy consumption. Similarly, people who report suffering from the cold during the previous winter, a proxy for energy poverty, do not increase their support for the AfD. If anything, it seems that people with higher perceived behavioral control over their energy consumption are more likely to increase their support for right-wing populists.

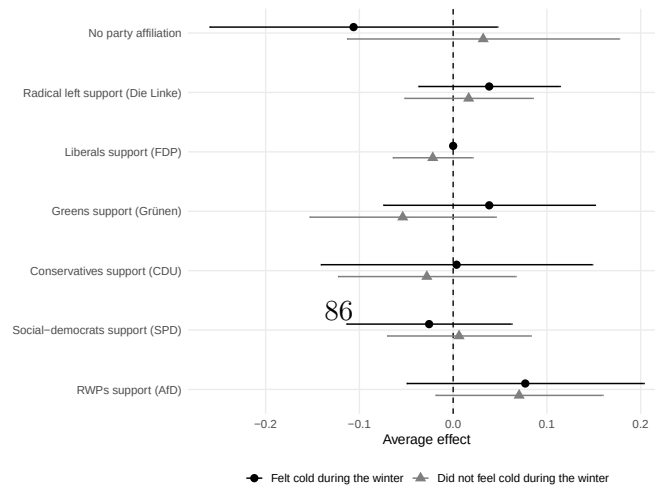
Figure 19: Unconditional D-i-D effects of receiving a higher electricity bill on partisan support: Split sample on energy poverty characteristics



(a) ATT of the binary treatment: 'Above median increase in electricity costs' among respondents with low/middle or high income



(b) ATT of the binary treatment: 'Above median increase in electricity costs' among respondents who state they can not/can easily reduce their energy consumption



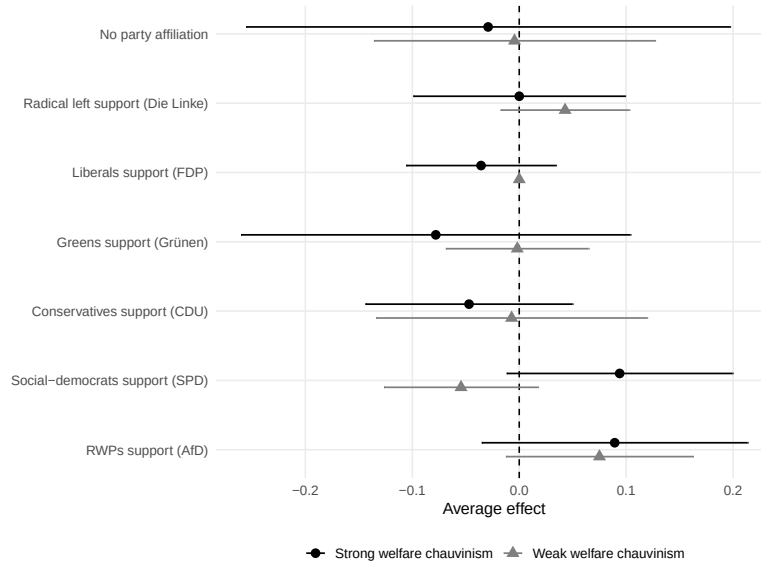
(c) ATT of the binary treatment: 'Above median increase in electricity costs' among respondents who reported feeling cold at home during the winter 2022-2023

Welfare chauvinism Another mechanism that could explain the rise of right-wing populists following the energy price shock is welfare chauvinism. Following [Guriev and Papaioannou \(2022\)](#), rising energy costs might exacerbate competition over limited state welfare support and employment opportunities. Under these pressures, voters prone to viewing immigrants as fiscal burdens or competitors on the labour market are particularly susceptible to AfD rhetoric linking energy and welfare scarcity to immigration-related fiscal stress, a concept known as welfare chauvinism. We expect to observe stronger increases in AfD support among voters with high baseline anti-immigrant attitudes. We measure such attitudes using the z-score of two survey items: 'Immigrants should have access to welfare state assistance to the same extent as all other German citizens.' and 'The federal government should give preferential treatment to German citizens when using public benefits.'³⁰

We do not find empirical patterns consistent with prior welfare chauvinism being driver of populist support. Figure 20 shows that the electricity costs shock has a similar effect among people with strong and weak baseline support for providing state welfare to immigrants.

³⁰The first item contributes negatively to the welfare chauvinism index.

Figure 20: Unconditional D-i-D effects of receiving a Above median increase in costs bill on partisan support, among respondents with high/low welfare chauvinism attitudes.



Attributional learning So far, our heterogeneity analysis does not convincingly answer the question: Why do some people experiencing a negative price shock increase their support for populist parties? Here, we suggest a new explanation to the political response to price shocks. The main mechanism is as follows: voters form beliefs on the capacity of mainstream parties to deliver *welfare-maximizing* policies, but cannot perfectly observe the policies. Instead, they revise their beliefs that mainstream parties implement welfare-maximizing policies when observing positive or negative utility shocks. If voters' trust in mainstream parties drops low enough, they turn to a populist party, who commits to enforce simple and specific policies (Morelli et al., 2021).

The assumption that voters are not fully aware of policy debates when forming political preferences builds on a long strand of literature in political economy. Coate and Morris (1995) assume voters cannot observe whether elected politicians are corrupted by private interests and the exact policies politicians implement. They show that this asymmetry of information gives rise to inefficient public spending. Maskin and Tirole (2004) assume that voters can eventually learn whether elected officials have implemented an optimal

policy and prove that the economic performance of public decision-making institutions strongly depend on the learning mechanism. [Persson et al. \(1997\)](#) show that the public can infer information about the efficiency of public funds by observing the interaction between the parliament and the executive government. The body of literature on political business cycles share a similar assumption, namely that voters do not perfectly observe expansionary policies prior to elections, which can be exploited by incumbents to increase chances of reelection ([Nordhaus, 1975](#); [Rogoff and Sibert, 1988](#); [Rogoff, 1990](#)). More recent empirical works have shown that the informational environment in which voters and politicians interact has a large effect on political outcomes ([Besley and Burgess, 2002](#); [Ferraz and Finan, 2008](#); [Snyder Jr and Strömberg, 2010](#)).

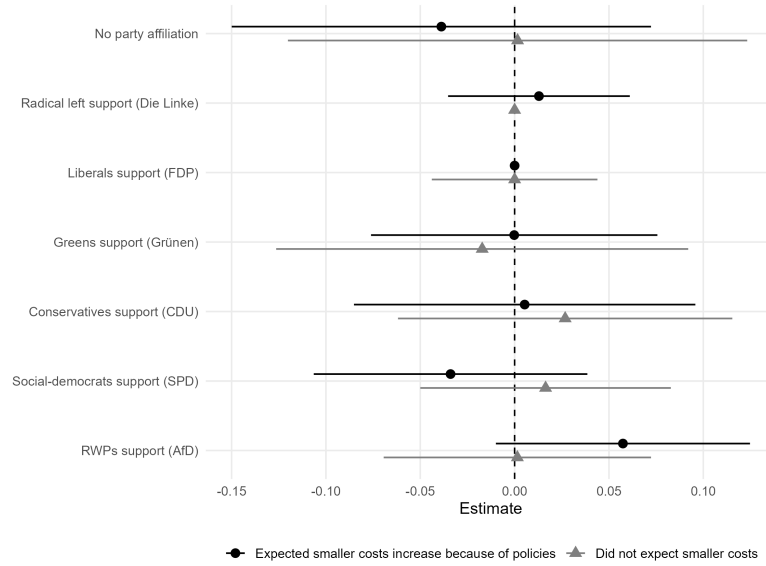
In general, this mechanism explains heterogeneous political responses to the energy price shock based on people's expectation of the signal. Intuitively, people will be more likely to vote for populists if their realised utility after the shock is smaller than the utility they were expecting, given their prior perceptions of the government policies.

We assess this prediction by comparing the effect of an increase in electricity instalment payments between respondents who, after receiving information about the electricity price brake (the main governmental compensation scheme, see Section 2.3), expected a 10% reduction in consumption to lower their electricity costs and respondents who expected no change. We expect an increase in electricity instalment payments to produce a larger increase in AfD support among respondents who expected savings from the policy than among those who expected no change, because the increase represents a larger negative utility surprise and induces a stronger downward revision in perceived mainstream policy quality.

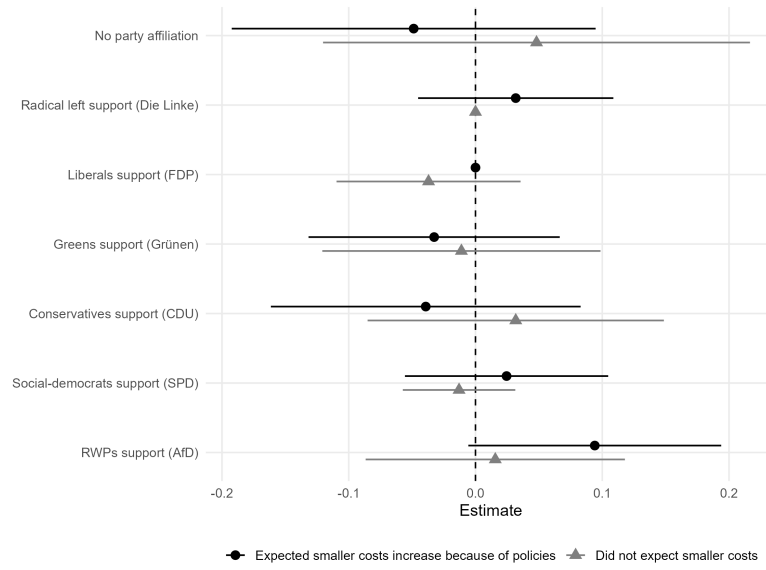
Consistent with the mechanism discussed above, we find that the electricity cost shock had a somewhat stronger effect on support for the populist party among those who expected government policies to decrease electricity costs (Figure 21).

This potential mechanism shares some features with [Callander and Carbajal \(2022\)](#). In [Callander and Carbajal \(2022\)](#), polarization is driven by citizens reducing the ideological gap to their favourite party, and partisan elites who strategically radicalise their voters.

Figure 21: Unconditional D-i-D effects of receiving a higher electricity bill on partisan affiliation among respondents who expected/did not expect a significant electricity costs reduction due to government compensation policy



(a) ATT of the binary treatment: 'Increased electricity costs'



(b) ATT of the binary treatment: 'Above median increase in electricity costs'

In our mechanism, polarization emerges not as a consequence of radicalisation in political ideology, but because of an epistemic disagreement between voters: people disagree on how to interpret economic outcomes, some blaming unfortunate utility realizations on bad policies, while others attribute them to bad luck.

Our mechanism would also be able to explain why shifts in political affiliation due to economic shocks are likely to persist over time, long after the material effects of the shock have gone. First, assuming committed populist parties do deliver on their promises after being elected, citizens would update their beliefs only if mainstream parties are able to convincingly signal that their policies would indeed be welfare-improving. If this is not the case, there would be few reasons for people to revise their belief that populist parties are preferable to mainstream parties. Second, an extreme negative economic shocks is likely to be followed by a positive one (simple regression to the mean). In the case a populist government arrives in power after the negative shock, they would be able to reap the political benefits from both negative and positive random economic shocks, thus reinforcing the perceptions that their policies are superior to those of the mainstream party.

E Party positions

Party competition over Russia’s war against Ukraine cannot be described adequately with a single conventional ideological scale. A party may support the admission of Ukrainian refugees while expressing greater caution over weapons. It may accept the economic costs associated with sanctions while opposing Ukraine’s membership of the European Union. Positions on the war may also coincide with established divisions over economic policy, cultural liberalism, European integration and anti-elite rhetoric. This section describes these relationships for Germany and assesses whether a smaller number of dimensions provides an informative account of the observed party positions.

E.1 Data and measurement

The analysis uses the March 2024 release of the Chapel Hill Expert Survey Ukraine module, which records party positions referring to 2023. The unit of observation is the party mean. The German sample comprises CDU, CSU, SPD, FDP, the Greens, Die Linke and AfD. The survey file contains no observation for BSW, which was established after the period covered by the expert assessments. Consequently, the results describe the party system represented in the survey and do not provide an estimate of the later position of BSW.

The survey supplies six dimensions concerning the response to the war. These measure support for admitting Ukrainian refugees, support for providing weapons, acceptance of higher energy costs arising from sanctions, support for Ukraine’s membership of the European Union, opposition to close relations with the Kremlin and the preferred strategy of the European Union towards Russia. Four further dimensions describe economic left–right position, GAL–TAN position, support for European integration and the salience of anti-elite rhetoric.

Most dimensions range from zero to ten. The measure of European integration ranges from one to seven. A higher value generally denotes greater support for Ukraine or European integration. Two qualifications are important. On the measure of strategy

towards Russia, zero denotes a preference for defence and security and ten denotes a preference for trade and diplomacy. The measure of anti-elite rhetoric records salience, with no implication that a high score denotes a policy position in the same sense as the other measures.

The analysis centres each dimension and divides it by its standard deviation before applying principal component analysis or hierarchical clustering. This procedure gives each dimension equal influence over the calculations despite differences in its original range and dispersion. No electoral weights enter these two procedures. Each party represents one observation because the purpose is to describe the configuration of the positions offered by parties.

The bivariate figures contain vertical and horizontal medians weighted by vote share. The weights come from the results of the 2021 German federal election in the 2024 stable release of ParlGov. The seven parties account for 91.25 per cent of valid votes and 735 of the 736 seats in the Bundestag. Medians weighted by seat share are identical to the medians weighted by vote share on all ten dimensions. The unweighted medians differ on the measures of refugees, European integration and the salience of anti-elite rhetoric. The dashed reference lines therefore describe the electoral centre among the observed party means. They provide no measure of uncertainty in the expert assessments.

E.2 The principal components

Figure 22 presents the first two principal components. The sign of the first component is determined by the loading for support for Ukrainian refugees. The sign of the second component is determined by the loading for economic left–right position. These conventions resolve the arbitrary sign of principal components and provide stable axis directions when the analysis is reproduced.

The first component accounts for 80.0 per cent of the standardised variation. Its positive side is associated with support for refugees, weapons, higher energy costs, Ukraine’s membership of the European Union, opposition to close relations with the Kremlin and support for European integration. The measure of strategy towards Russia has a neg-

ative loading. This association arises from its original direction, in which higher values indicate greater emphasis on trade and diplomacy. The salience of anti-elite rhetoric also has a negative loading. The first component consequently distinguishes the parties that support a broad response to the invasion from AfD and Die Linke, although the two latter parties differ considerably on the conventional ideological dimensions.

The second component accounts for a further 16.9 per cent of the standardised variation. Its largest positive loadings concern economic right position and the traditional or authoritarian side of the GAL–TAN scale. CDU, CSU, FDP and AfD receive positive scores on this component. SPD, the Greens and Die Linke receive negative scores. The first two components account for 96.9 per cent of the standardised variation overall. The plane displayed in Figure 22 therefore retains nearly all of the variation among the seven party profiles.

The loading vectors in panel (b) clarify the substantive content of the two components. The measures of weapons, energy costs, Ukraine’s membership of the European Union and opposition to close relations with the Kremlin point in similar directions. Support for refugees and support for European integration belong to the same broad group. Economic left–right position and GAL–TAN position contribute most strongly to the second component. Strategy towards Russia and the salience of anti-elite rhetoric point towards the negative side of the first component. This pattern explains why AfD and Die Linke are located on the same side of the first component even though their economic and cultural positions remain distinct.

E.3 Dimensionality and hierarchical clustering

Figure 23 reports the distribution of variation across the components and the results from Ward hierarchical clustering. Seven parties permit at most six components with positive variance. The first component accounts for four fifths of the variation and the second raises the cumulative proportion to 96.9 per cent. The remaining components add little descriptive information. This concentration supports the use of the first two components as a concise representation of the observed party profiles.

The clustering procedure uses Euclidean distances between the standardised positions and Ward’s criterion for the successive mergers. CDU and CSU form the closest initial pair, after which FDP joins them. SPD and the Greens form a second pair. Die Linke and AfD form a separate branch, while their merger occurs at a greater linkage height than the mergers within the other branches. The final merger separates CDU, CSU, SPD, FDP and the Greens from Die Linke and AfD.

The proximity of Die Linke and AfD in the dendrogram describes similarity across all ten standardised values. It does not imply ideological equivalence. Both parties express relatively weak support for several responses to the war and assign greater salience to anti-elite rhetoric. They remain far apart on economic policy and GAL–TAN position. Ward clustering is sensitive to the dimensions included and to their standardisation, which reinforces the need to consult the underlying values.

Figure 24 displays the ten original party means and their standardised values. The number in each cell retains the original survey scale. Cell colour reports the corresponding standardised position across the seven parties. The order of the rows and columns follows Ward clustering, which places similar profiles next to one another.

The heatmap confirms a broad group of parties with comparatively strong support for Ukraine. CDU, CSU and FDP receive high scores on weapons, opposition to close relations with the Kremlin and acceptance of the economic costs associated with sanctions. The Greens combine high support across these measures with the highest support for refugees and Ukraine’s membership of the European Union. SPD is close to this group on refugees and European integration, with more moderate positions on weapons and the economic costs of sanctions.

AfD occupies the opposite end of most dimensions concerning Ukraine. It receives the lowest scores on weapons, energy costs, Ukraine’s membership of the European Union and opposition to close relations with the Kremlin. Its high score on strategy towards Russia denotes a preference for trade and diplomacy. It also combines a traditional or authoritarian GAL–TAN position with very high salience of anti-elite rhetoric. Die Linke also receives low scores on weapons and acceptance of higher energy costs, and it favours

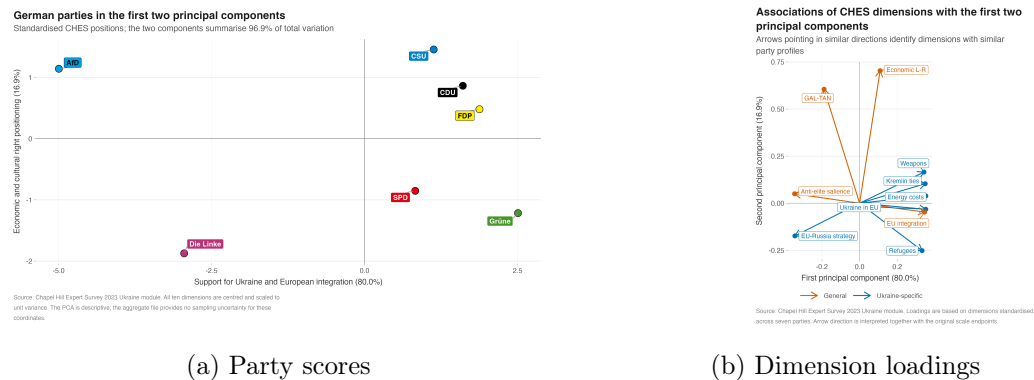


Figure 22: German parties in the space defined by the first two principal components. *Notes.* Panel (a) shows party scores and panel (b) shows the corresponding dimension loadings. All ten dimensions are centred and divided by their standard deviation. The first component accounts for 80.0 per cent and the second component accounts for 16.9 per cent of the standardised variation. Party labels use official colours. *Source.* Chapel Hill Expert Survey 2023 Ukraine module.

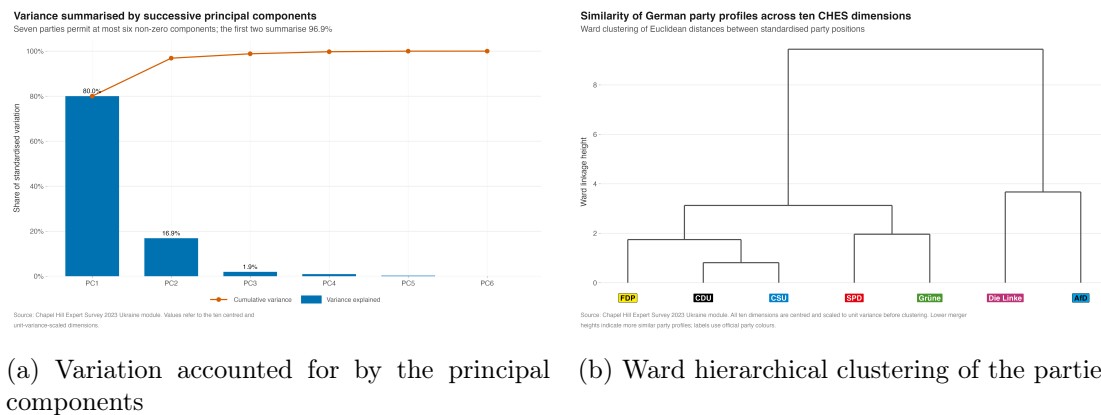


Figure 23: Dimensionality and similarity among German party positions. *Notes.* Panel (a) reports the proportion of standardised variation associated with each principal component and its cumulative proportion. Panel (b) reports Ward hierarchical clustering based on Euclidean distances between the same standardised party positions. Lower linkage heights indicate more similar profiles. *Source.* Chapel Hill Expert Survey 2023 Ukraine module.

trade and diplomacy in relations with Russia. Its economic position remains strongly on the left and its GAL–TAN position is much less traditional or authoritarian than that of AfD.

E.4 Selected comparisons between dimensions

The principal components provide a concise account of common variation, while the selected bivariate comparisons preserve the substantive scales. Figure 25 considers four measures that cover humanitarian assistance, military assistance, economic costs and Ukraine’s membership of the European Union.

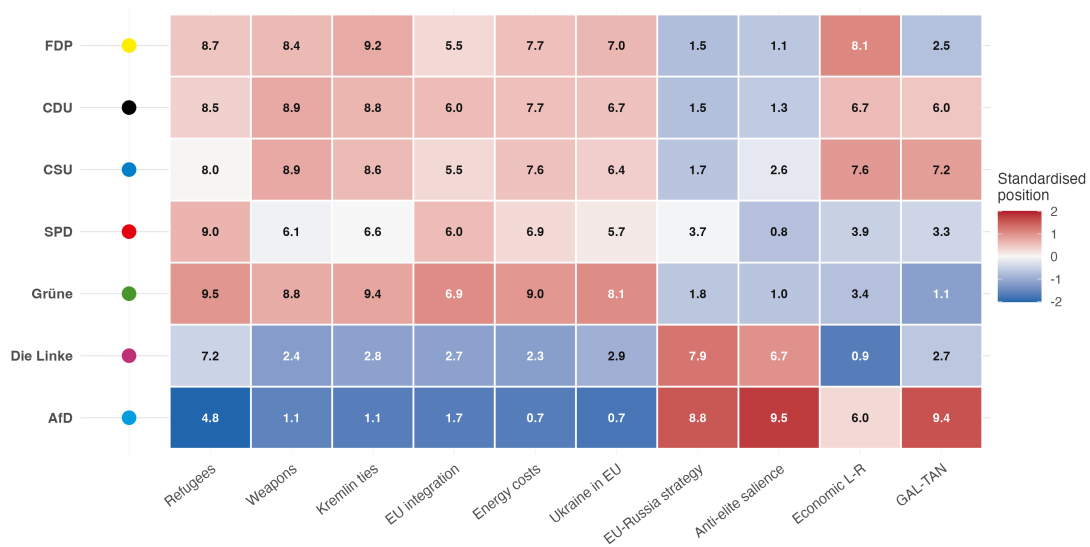
Panel (a) shows a clear distinction between support for refugees and support for weapons. The Greens, CDU, CSU and FDP combine high scores on both measures. SPD receives a high score on refugees and a more moderate score on weapons. Die Linke and AfD are located well below the median weighted by vote share on weapons, while AfD also receives the lowest score on refugees. The separation between Die Linke and AfD on refugees demonstrates that their proximity on the first principal component does not extend to every aspect of policy towards Ukraine.

Panel (b) compares acceptance of higher energy costs with support for Ukraine’s membership of the European Union. The two positions correspond closely. The Greens occupy the upper right, followed by FDP, CDU and CSU. SPD receives moderately supportive scores on both dimensions. Die Linke and AfD are below the medians weighted by vote share on both dimensions. The association indicates that acceptance of the domestic economic consequences of sanctions coincides with support for a longer institutional commitment to Ukraine among the parties in this survey.

Figure 26 places the parties on the four general dimensions. Panel (a) shows that economic and cultural positions provide distinct information. FDP has the most economically liberal position and remains on the libertarian or postmaterialist side of the GAL–TAN scale. CSU combines an economically right position with a more traditional or authoritarian position. AfD is less economically right than FDP or CSU and is by far the most traditional or authoritarian party. SPD, the Greens and Die Linke remain on

Clustered profiles of German parties across CHES dimensions

Rows and columns are ordered by Ward hierarchical clustering; cells display the original party means



Source: Chapel Hill Expert Survey 2023 Ukraine module. Colour shows within-dimension z-scores across seven parties; numbers retain the original scales. High values follow each source dimension's documented direction. Party markers use official colours.

Figure 24: Clustered profiles of German parties across ten CHES dimensions. *Notes.* Rows and columns follow Ward hierarchical clustering. Cell colour reports the position after centring each dimension and dividing it by its standard deviation. Numbers report the original party means. Higher values retain the documented direction of each source measure. Party markers use official colours. *Source.* Chapel Hill Expert Survey 2023 Ukraine module.

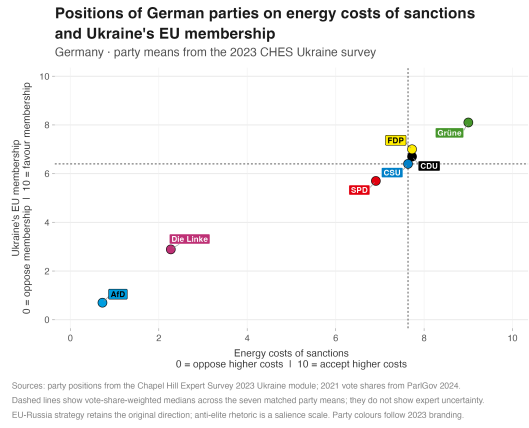
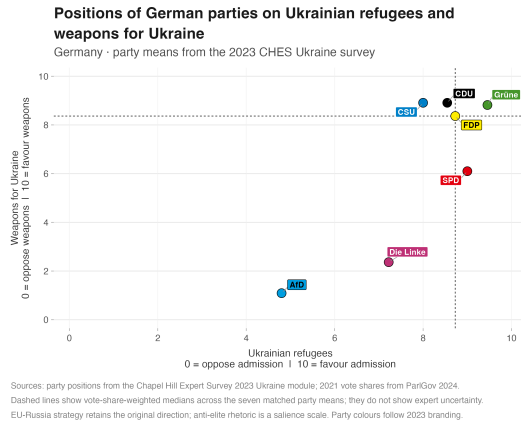
the economic left, while the Greens receive the lowest GAL–TAN score.

Panel (b) displays an especially clear distinction between support for European integration and the salience of anti-elite rhetoric. CDU, SPD, FDP, the Greens and CSU combine support for European integration with low salience of anti-elite rhetoric. Die Linke receives a lower score on European integration and a substantially higher score on anti-elite rhetoric. AfD receives the lowest score on European integration and the highest score on anti-elite rhetoric. This comparison helps to explain the strong contribution of these two measures to the first principal component.

E.5 Interpretation and limitations

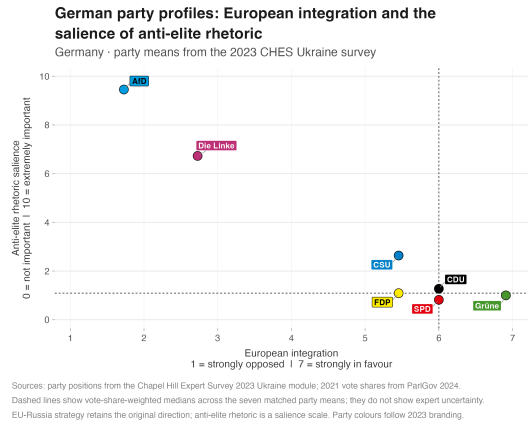
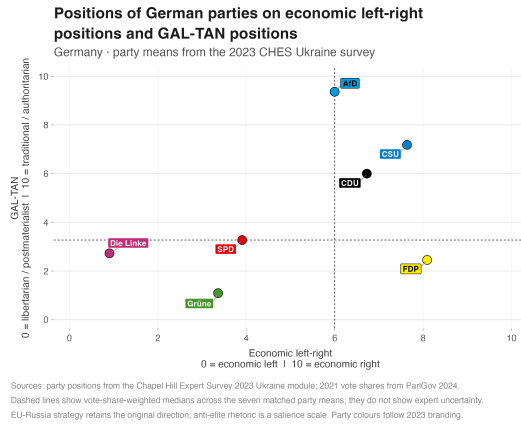
Overall, the German party positions contain one dominant dimension concerning support for Ukraine and European integration, accompanied by a second dimension that reflects economic and cultural ideology. The first dimension separates CDU, CSU, SPD, FDP and the Greens from AfD and Die Linke. The second distinguishes parties on the economic and GAL–TAN scales and prevents the proximity of AfD and Die Linke on policy towards Ukraine from being construed as general ideological similarity.

The results remain descriptive. Seven observations provide a narrow basis for estimating a more general latent structure, and the aggregate CHES file supplies no standard errors for the party means. The component scores and clusters describe the parties included in the 2023 expert assessments. They do not estimate voter preferences, changes after the survey or the position of BSW. The medians in the bivariate figures depend on the 2021 election results, although the medians weighted by vote share and seat share coincide on all ten dimensions. The complete collection of 45 bivariate comparisons is therefore best retained as supplementary material, while the five figures reported here provide the principal evidence required for the appendix.



(a) Ukrainian refugees and weapons for Ukraine (b) Energy costs of sanctions and Ukraine's membership of the European Union

Figure 25: German party positions on support for Ukraine and associated policy commitments. *Notes.* Dashed lines report medians weighted by vote share across the seven observed parties. The weights come from the 2021 German federal election in the 2024 stable release of ParlGov. The lines do not represent uncertainty in the expert assessments. Party labels use official colours. *Sources.* Party positions come from the Chapel Hill Expert Survey 2023 Ukraine module. Electoral weights come from ParlGov.



(a) Economic left-right and GAL-TAN positions (b) European integration and the salience of anti-elite rhetoric

Figure 26: German party positions on conventional ideological dimensions. *Notes.* Dashed lines report medians weighted by vote share across the seven observed parties. Economic left-right and GAL-TAN positions use scales from zero to ten. European integration uses a scale from one to seven. The salience of anti-elite rhetoric uses a scale from zero to ten. Party labels use official colours. *Sources.* Party positions come from the Chapel Hill Expert Survey 2023 Ukraine module. Electoral weights come from the 2024 stable release of ParlGov.

F Theoretical model

F.1 Party choice under material shock

We formalise the argument in Section 5.1 as follows. Consider a policy space with $K \in \mathbb{N}$ dimensions – for example immigration, redistribution, climate policy, foreign policy, and support for democratic institutions. Let $x_{i0} \in \mathbb{R}^K$ denote voter i 's ideal point prior to the shock, and $p_j \in \mathbb{R}^K$ party j 's position. The voter's spatial distance from party j is

$$D_{ij} = \sum_{k=1}^K \omega_{ik} (x_{ik,0} - p_{jk})^2, \quad \omega_{ik} \geq 0, \quad (11)$$

where ω_{ik} is the weight voter i places on dimension k .

Let h_i denote the instalment increase experienced by household i , t_i its timing, and $r_i(t_i)$ the compensation available at that date, so that the net loss is $z_i = h_i - r_i(t_i)$. Let R_i denote household resources over the relevant horizon and q_{ij} the present value of energy costs, net of expected compensation, that voter i expects under party j – beliefs about wasteful government spending raise q_{ij} by lowering the expected value of future compensation. Electoral utility from supporting party j is

$$V_{ij} = u(R_i - z_i - q_{ij}) - D_{ij}, \quad u' > 0, \quad u'' < 0, \quad (12)$$

and the voter supports $j_i^* \in \arg \max_{j \in \mathcal{C}_{i,t_i}} V_{ij}$, where $\mathcal{C}_{i,t_i}$ is her consideration set at the time of exposure.

Writing A for the AfD: if $q_{iA} < q_{ij}$ for her current party j , a larger z_i raises $V_{iA} - V_{ij}$ under $u'' < 0$, and the voter switches from j to A once

$$u(R_i - z_i - q_{iA}) - u(R_i - z_i - q_{ij}) \geq D_{iA} - D_{ij}. \quad (13)$$

For a fixed spatial gap $D_{iA} - D_{ij}$, the switch occurs only once the loss z_i is large enough for the material advantage of the AfD to clear that gap.